Is Real Madrid special? Testing the claim that VAR works against Real Madrid with nine seasons of La Liga data

Data mining, cleaning, validation, statistical framework and results

Draft for review · 12 October 2026 · Revised 15 October 2026 (pipeline audit added as Section 4.5 and Appendix D; no estimate changed) · Scope: La Liga, 2018-19 (VAR introduced) to 12 October 2026 (2026-27 partial, 8 rounds) · Code and data: github.com/RoloMagnus/Real_Madrid_VAR_null_goals_outlier_study

Abstract. On 10 October 2026 a widely shared post listed the La Liga teams with the most goals ruled out by VAR since 2018; Real Madrid led with 42 (+1 that day), nearly twice the runner-up. Fans read this as evidence of referee or VAR bias against the club. We (i) rebuilt the underlying event data from four public sources covering all 3,118 league matches played since VAR was introduced, (ii) validated the post and our own reconstruction against independent providers and a manual video check, and (iii) tested whether the excess is explained by how much and how Real Madrid attacks, or whether it carries the fingerprints of unequal treatment.

The outlier is real but the post's exact figures are not reproducible: we find **50** VAR-nullified goals for Real Madrid versus 30 for the next team (Villarreal), a rate of 0.160 per match against 0.058 for all other teams (rate ratio 2.8, 95% CI 2.0–3.7). Attack volume, offside volume, pressing, fouls and cards together explain about half of the count (a model fitted on the other teams predicts ~25 goals, 95% interval 16–35); a residual rate ratio of about 2.0 remains. However, the residual shows none of the signatures of targeting: within the same matches the split of nullified goals follows each side's share of offsides and goal opportunities, opponents are not significantly spared, Real Madrid's rate does not differ across 15 referees, the same excess appears in goals ruled out on the pitch without any VAR review, and published VAR accuracy (96–98%) is far too high for errors to account for it. Counting *every* Real Madrid nullified goal as wrong would change one league title (2020-21). The unexplained residual is most consistent with Real Madrid's attacking profile, a hypothesis that requires offside-line tracking data to confirm.

Headline results

- 50 VAR-nullified Real Madrid goals in 44 of 312 league matches; 15 of those matches ended in a draw or loss and in 13 the result would have changed (21 points over 8.2 seasons).
- Rate ratio versus other teams: 2.76 raw → 2.01 after controlling for offsides, shots, xG, home and season (top-10 teams only: 1.84); 1.99 with pressing (PPDA) added; 2.02 with fouls and cards added.
- No significant within-match asymmetry, no referee concentration, no evidence that opponents are protected; on-field (non-VAR) disallowed goals show the same direction (rate ratio 2.0); the extra Real Madrid effect at the VAR stage is 1.36× (95% CI 0.82–2.23, p = 0.23).
- Title counterfactual (all 50 goals counted): 1 of 5 non-title seasons flips (2020-21); at a realistic 1.7–5% VAR error rate the chance is 1.9%–5.1%.

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1. Background and research question

The Video Assistant Referee (VAR) was introduced in La Liga in the 2018-19 season. On 10 October 2026 the analyst MisterChip (@2010MisterChip) published a table titled *"Equipos con más goles anulados por VAR desde que se usa el videoarbitraje en La Liga (2018-2026)"*: Real Madrid 42 (+1 that day), Atlético de Madrid 22, Villarreal 22, Celta 20, Real Sociedad 20, Sevilla 20 and Barcelona 19. The post reached 1.1 million views and prompted claims of referee influence and institutional corruption against Real Madrid.

A ranking of raw counts cannot support such claims by itself. Counts depend on exposure (how often a team attacks and scores), on the way it attacks (a team that plays runners behind a high line produces more offside goals), on the data source, and on chance. We therefore ask four questions:

1. **Validation.** Are the figures in the post correct?
2. **Description.** For each Real Madrid goal ruled out after VAR review: when, against whom, by which player, why, under which referee and at what position in the table?
3. **Inference.** Is Real Madrid's count an outlier after accounting for attacking volume and style, and, if so, does the residual carry the fingerprints of *targeting* (against Real Madrid) or *favouritism* (towards its opponents)?
4. **Impact.** Did these decisions cost Real Madrid points, and would any of them have changed a league title?

1.1 Hypotheses

- **H0 (neutral officiating).** A team's nullified goals are Poisson-distributed with a rate depending only on its attacking exposure; Real Madrid has no special effect.
- **H1a (targeting or favouritism).** Real Madrid's rate exceeds what exposure implies *and* the excess appears within matches (same referee, same VAR team) as opponents being spared, and/or concentrates in specific referees, and/or is specific to the VAR stage.
- **H1b (profile).** Real Madrid's rate exceeds what simple exposure implies, but the excess is symmetric within matches, spread over referees and equally present in decisions taken without VAR: a property of how the team attacks rather than of how it is treated.

The analysis is observational. It can reject specific, observable consequences of H1a; it cannot prove the absence of bias.

2. Data collection

No official VAR log exists in the public domain. Event data were therefore assembled from four public sources, each used for what it does best, with deliberate redundancy so that no conclusion depends on a single provider (Figure 1).

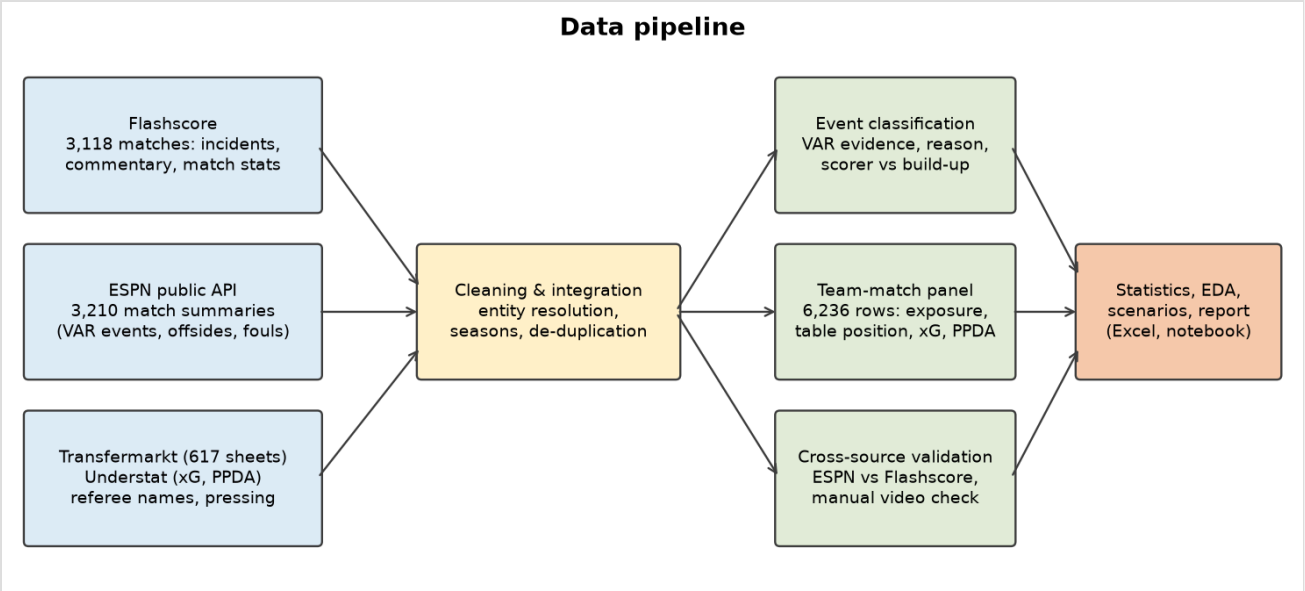


Figure 1. Data pipeline. Three families of public sources are harmonised, classified, cross-validated and then analysed.

Table 1. Data sources. All are public, unofficial endpoints queried programmatically and cached locally.

Source	Content used	Volume	Role
Flashscore (public match feeds)	Per match: incidents (including *Goal Disallowed* with scorer, minute, reason), live commentary (VAR wording, referee names), team statistics (shots, shots on target, offsides, corners, fouls, cards, possession)	3,118 matches, 2018-19 to 12-Oct-2026	Primary source of nullified goals and exposure variables
ESPN public API	Match summaries: play-by-play commentary with typed events (*Offside*, *Deleted After Review*, *VAR decision*), box score, rosters	3,210 summaries (3,112 completed)	Independent corroboration of events; offside player names; foul locations
Transfermarkt	Match sheets with the full name of the referee	617 sheets (every match with a nullified goal plus all Real Madrid matches)	Referee identity
Understat	Team-match expected goals (xG) and PPDA components	6,234 team-matches (all seasons)	Chance quality and pressing covariates

2.1 Acquisition

ESPN. The scoreboard endpoint accepts a single day or a calendar year but rejects date ranges. Nine yearly requests (2018–2026) enumerate all fixtures, which were assigned to seasons by date (August–July, with the COVID-delayed 2019-20 finish in June–August 2020 mapped explicitly). Each match summary was then downloaded concurrently (12 threads, 4 retries, on-disk cache): 3,210 summaries, of which 3,112 were completed matches, 97 scheduled fixtures and one abandoned match.

Flashscore. Each season's results page embeds the first 25 matches; further pages are available from a paginated feed keyed by a tournament-season code, which we discovered by inspecting the page's JavaScript and scanning candidate codes. For each of the 3,118 completed matches we downloaded four feeds (incidents, statistics, match information with the referee, and live commentary). Commentary for every match was downloaded because the incident feed alone omits goals chalked off by on-field officials. Season coverage is complete: 380 matches in each of the eight full seasons plus 78 in 2026-27.

Transfermarkt and Understat. Matchday pages were scraped to map each fixture to its match-sheet identifier (3,109 of 3,118 matched after an alias fix for Athletic Club/Athletic Bilbao), then the sheets

needed for the analysis were fetched. Understat's league endpoint returned team-match xG and PPDA for every season.

Sources that failed. Sofascore returned HTTP 403 for scripted requests and a search engine returned a captcha; both were abandoned rather than circumvented. ESPN's referee field was populated for only 116 matches and was therefore not used.

Terms of use. The endpoints are undocumented and unofficial; the terms of the providers apply. The repository does not redistribute the raw caches (about 1.6 GB, regenerable by the scripts); it contains processed, derived tables.

3. Data cleaning and integration

3.1 Entity resolution

The same club appears under different names in each provider (e.g. *Atl. Madrid*, *Atlético Madrid*, *Atlético de Madrid*; *Ath Bilbao*, *Athletic Club*; *Celta Vigo*, *Celta de Vigo*). Names were normalised by stripping diacritics, lower-casing and token-set matching, with a small explicit alias table for the residual cases. Players are given as *Mbappe K.* in Flashscore; we reconstructed full accented names by converting Flashscore's URL slug (`mbappe-kylian`) to a token set and looking it up in ESPN's roster names (*Kylian Mbappé*).

3.2 Time, seasons and match keys

Kick-off times were converted from UTC to Europe/Madrid. ESPN and Flashscore matches were joined on (kick-off hour, home goals, away goals), with ambiguous joins dropped rather than guessed; Flashscore's own identifier is the primary key of the panel. A post-draft audit (Section 4.5) found that the foul-zone join, which used a looser team-name match, attached ESPN's La Coruña–Valencia match to the Alavés–Valencia fixture of September 2026 for the foul-zone covariates; this affects one match and no result.

3.3 Defining the outcome: a VAR-nullified goal

A goal is **VAR-nullified** if it was scored, then ruled out after a VAR review. Flashscore's incident feed logs 397 *Goal Disallowed* events league-wide (offside 281, foul 50, handball 49, unspecified video review 17). An event is **VAR-confirmed** if either (a) Flashscore commentary within −1/+3 minutes carries VAR wording (the explicit *var* icon, or text such as "VAR review", "video assistant referee"), or (b) ESPN logs an overturned goal, a VAR "no goal" decision or a "Video Review" delay within three minutes. This gave **394** VAR-confirmed events; three events with no VAR evidence (none involving Real Madrid) were excluded from all tests.

Events were then classified: by **reason** (offside, attacking foul, handball, unspecified) and, for offsides, by **who** was offside using ESPN's typed offside events (a matching event within ±1 minute for the same team): *goalscorer* (125 of 397 events), *teammate in the build-up* (101) or *not identified* (57). The latter is how the "passive player" field of the report is filled. The half is derived from the base minute (stoppage time inherits the half). The **context** of each goal (score before the goal, what the goal would have meant, final result, points that would have been gained) is computed from the incident feed's running score.

> **A boundary case that shaped the definition.** During manual review a second Mbappé goal in the 0–4 Clásico of 26 October 2024 (65') was proposed. It is genuine ("The assistant referee raises his flag and the referee blows his whistle. Mbappé puts the ball into the net, but he is flagged for offside") but involved no VAR review in either feed. It was therefore not a VAR-nullified goal. This motivated a separate dataset of

> goals ruled out *on the pitch* (Section 3.7) and a test of whether the VAR stage differs from the on-field stage (Section 6.6).

3.4 Referees

Full referee names come from Transfermarkt match sheets for all 617 relevant matches (100% resolved for matches with a nullified goal); Flashscore's abbreviated names were mapped to full names by majority vote over paired matches. There are 41 distinct referees. The referee recorded is the **on-field referee**, the formal decision-maker after VAR advice; the VAR-room official is not published consistently by any accessible source.

3.5 League position at the time of each goal

No provider supplies the table at kick-off, so every season was rebuilt from results. For the panel, the table before each kick-off is sorted by points, goal difference, goals scored. Round-1 events have no table and are left blank (7 events). For the title scenarios (Section 6.8) the full La Liga tie-break order is applied (points, head-to-head points, head-to-head goal difference, overall goal difference, goals scored). The reconstruction was checked against known outcomes: the champion and top four of each season match the earlier table, and champions' points match the official totals (e.g. Barcelona 87, Real Madrid 87, Atlético 86, Real Madrid 86, Barcelona 88, Real Madrid 95, Barcelona 88 and 94).

3.6 Exposure and covariate construction

- **Shots, shots on target, offsides, corners, fouls, cards, possession** from Flashscore match statistics. Red cards are not listed when zero (79% blanks) and were set to zero.
- **xG and PPDA** from Understat, joined by team and date (±2 days). Flashscore xG exists only from 2022-23 and was used as a fallback and as a cross-check. 1.2% of team-matches lack xG and 1.9% lack PPDA, almost entirely Huesca and Oviedo, whose names differ between Flashscore and Understat ("Huesca" vs "SD Huesca", "Oviedo" vs "Real Oviedo"; see Section 4.5); models are fitted on complete cases and the sample size is reported with each model. PPDA is passes allowed per defensive action; the opponent's PPDA in the same match is taken from the other team's row.
- **Fouls by zone.** ESPN records a foul as two lines: *Foul by X (Team)* and *Y (Team) wins a free kick in the defensive half / attacking half / on the left (right) wing*. The location is stated from the fouled player's perspective, so a free kick won "in the defensive half" is a foul committed in the committer's *attacking* half. We verified the orientation against player roles (forwards generate that tag 78% of the time, defenders about 44%). Penalty-area fouls are logged only as *Penalty conceded by…* and were parsed separately. ESPN captures about 91% of box-score fouls, so zone counts are used as shares and as covariates, not as totals.
- **Top-10 scope.** At the request of the study sponsor, comparisons are also reported for the ten best teams of each season's final table, which removes relegated teams by construction (see threat 3 in Section 8).

3.7 Goals ruled out on the pitch

Mining the commentary of all matches for five recurring phrasings ("puts the ball into the net, but he is flagged for offside"; "A goal, which was scored by…, was disallowed due to…"; "Goal disallowed! X scores but the referee blows his whistle for a foul"; "immediately ruled out by referee… due to a foul/handball"; "thought he had scored but it was disallowed…") and removing lines mentioning VAR, events within four minutes of a VAR-confirmed event for the same team, and duplicate descriptions of the same goal yields **298** on-field disallowed goals (Real Madrid 29). Because it relies on phrasing, this count is a floor.

3.8 Resulting analysis panel

The analysis panel has 6,236 team-matches (3,118 matches × 2), each with exposure covariates, table position, VAR-nullified and on-field disallowed counts, and the season's final position. The Real Madrid breakdown has 50 rows (one per goal) with 28 fields.

4. Validation

4.1 Reconstructing the post

The post's figures are lower than our reconstruction for every team listed (Table 2, Figure 2). Real Madrid is first in both, and the ranking of the top is similar, but the post's runner-up gap (42 vs 22, 1.9×) is wider than ours (50 vs 30, 1.7×), and Barcelona is third in our data but seventh in the post. Restricting to offside and attacking-foul reasons reproduces Real Madrid (42), Atlético (22) and Celta (20) exactly but not the other teams, so no single definition reconciles the post. We conclude that the post's provider and definition are unknown, that its exact numbers should not be relied upon, and that its *direction* (Real Madrid first, by a wide margin) is confirmed.

Table 2. The post versus the reconstruction (La Liga, 2018-19 to 9-Oct-2026). The post counts one additional Real Madrid goal on 10 October (our figure for all dates: 50).

Team	Post	Reconstruction (all VAR-confirmed, before 10-Oct)	Reconstruction (offside + attacking foul)	Difference vs post (%)	Rank in reconstruction (all teams)
Real Madrid	42	49	42	17	1
Atlético de Madrid	22	26	22	18	4
Villarreal	22	30	27	36	2
Celta de Vigo	20	23	20	15	6
Real Sociedad	20	21	18	5	7
Sevilla	20	25	19	25	5
FC Barcelona	19	27	25	42	3

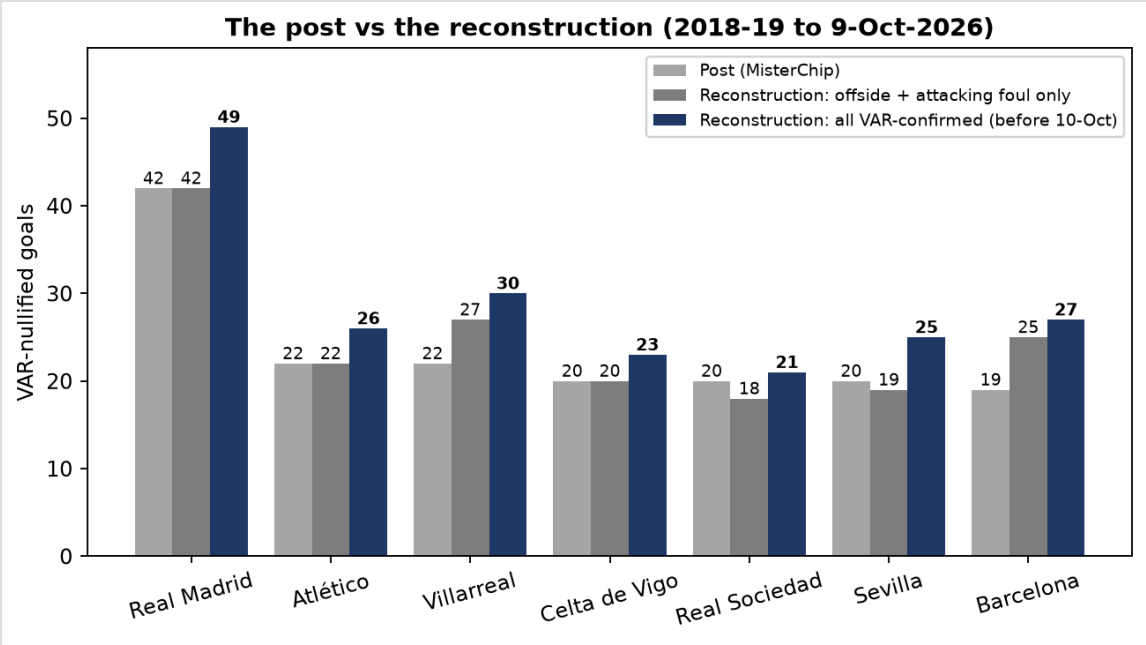


Figure 2. The post compared with two reconstructions. The narrow definition reproduces three teams exactly; no definition reproduces all seven.

4.2 Cross-provider agreement of exposure variables

Team-level statistics from Flashscore were compared with ESPN box scores for 6,068 team-matches (Table 3). Shots, shots on target, offsides and corners agree exactly in 96–98% of cases and within one unit in 98–99%. Flashscore xG (available 2022-23 onwards) correlates at 0.93 with Understat's xG, which was used throughout.

Table 3. Agreement between Flashscore and ESPN box-score statistics (xG: Flashscore vs Understat).

Metric	Team-matches compared	Pearson r	Exact match (%)	Within +/-1 (%) (xG: within 0.25)
Total shots	6,068	0.964	95.7	98.5
Shots on target	6,068	0.979	96.8	98.7
Offsides	6,068	0.980	98.4	99.1
Corners	6,068	0.975	97.1	98.8
Fouls	6,068	0.936	97.9	98.6
xG (Flashscore vs Understat)	2,794	0.925		60.7

4.3 Corroboration of events by an independent provider

ESPN only began tagging overturned goals explicitly in 2022-23, so it cannot serve as ground truth for the full period. For 2023-24 onward its explicit “GOAL OVERTURNED BY VAR” events rank Real Madrid first with 16 goals (Flashscore: 21) and a 12.7% share of league-wide events (Flashscore: 12.2%): the outlier does not depend on one provider's logging habits (Table 4). Corroboration of Flashscore events by any ESPN VAR evidence (explicit event, no-goal decision or review delay within three minutes) ranges from 73% to 94% by season; unmatched events are mostly ones ESPN does not label, not contradictions.

Table 4. ESPN's own overturned-goal log versus Flashscore, 2023-24 to 2026-27 (top 8 teams).

Team	ESPN "overturned by VAR"	Flashscore nullified	ESPN share (%)	Flashscore share (%)
Real Madrid	16	21	12.7	12.2
Villarreal	13	19	10.3	11.0
Atlético de Madrid	12	15	9.5	8.7
Celta de Vigo	11	14	8.7	8.1
FC Barcelona	8	14	6.3	8.1
Real Sociedad	10	10	7.9	5.8
Sevilla	5	10	4.0	5.8
Espanyol	7	9	5.6	5.2

4.4 Manual validation

A hand-built sample of four rows, prepared by the study team independently of the pipeline, was checked against the raw feeds. **The errors it exposed were errors of the hand-built sample, not of the pipeline output**: one wrong minute (18', not 43'), one match with no VAR review at all (Valencia, 1 December 2018), and one row that merged two matches (the Modrić/Varane offside belongs to the 16 March 2019 match against Celta, not the Valladolid match). The pipeline already held the correct record for each: Real Madrid–Levante at 18'; no incident at all for Valencia on 1 December 2018; Modrić's goal in the Celta match; and only Valladolid's own goal in the Valladolid match. Such errors show how bad state can enter a manually assembled table, which is why Section 4.5 audits the pipeline itself. All 50 Real Madrid goals were then reviewed by the study team against match video. The final breakdown matches the manually validated list one-to-one (50 of 50, matched on date, player and minute) and carries the validation flag and video link of each goal.

4.5 Pipeline audit (added after publication of the first draft)

A reader observed that a clean pipeline can still carry bad state all the way through, citing the kinds of error found in the hand-built sample: a wrong minute, a non-existent VAR review and merged matches. We therefore tested each type directly against the raw feeds for the whole dataset (scripts in `scripts/audit/`; full write-up in `AUDIT.md`; summary in Appendix D).

- **Minutes.** Flashscore's incident minute is the minute of play in which the goal was scored (1-indexed; a goal at clock time 53:43 is the 54th minute). ESPN stamps the VAR outcome at decision time. For the 286 events that ESPN tags explicitly, the two minutes are identical in 47% of cases, within one minute in 79% and within two in 95.5% (all found inside a ±3-minute search window); Flashscore is earlier in 136 events because the decision follows the review. Differences of one minute against other sources are a rounding convention, not an error.
- **VAR reviews that never happened.** An independent re-derivation of the evidence behind all 397 candidate incidents found explicit VAR wording and/or an explicit ESPN VAR event for 391 (286 have both). Of the other six, three were excluded; two are counted on an ESPN “Video Review” delay alone (Real Madrid–Levante 2018, validated against video by the study team, and Valencia–Girona 2018); and **one is counted although its commentary never mentions VAR** (Getafe, July 2020, not a Real Madrid goal), because the matching rule accepted the bare word “disallow”, which also describes on-field decisions. In a stricter dataset that keeps only explicit VAR wording (391 events, Real Madrid 49) the rate ratio is 2.72 (95% CI 1.97–3.68) against 2.76 in the main analysis. In all 389 events with VAR commentary the VAR lines refer to the goal (none concerns only a penalty or a card), no incident is counted twice, and 122 of the 128 goals ESPN labels as overturned by VAR from 2022-23 onwards are in our list (none of the six missing involves Real Madrid).

- **Merged or mismatched matches.** In the main pipeline: none. For the 3,034 matches with exactly one ESPN candidate under the kick-off-hour-plus-score key, team names agree in all 3,034; the 78 ambiguous matches are left blank by design (18 events, none Real Madrid). In all 6,120 team-matches joined to Understat the goals scored and conceded equal the Flashscore score, and Transfermarkt referee surnames agree with Flashscore in 616 of 617 matches (one 2026-27 disagreement, not a Real Madrid match). Two defects were found in secondary covariates: the foul-zone script mapped ESPN's "Deportivo" (La Coruña) to Deportivo Alavés, so one fixture received another match's foul data and seven La Coruña matches have none; and the Huesca and Oviedo aliases were missing, which removed 76 team-matches from every xG model and 230 from the PPDA models.

In total the audit found five small defects (Appendix D, Table 24). None changes Real Madrid's count of 50, and re-running the analysis with the defects fixed in an isolated copy moves no estimate by more than 0.05 on a rate ratio of about 2.0 (Table 25). We left the published estimates unchanged and document the fixes. A more fundamental limitation remains: all sources are unofficial public feeds and Flashscore and ESPN are not guaranteed to be independent of each other (both may derive from the same upstream provider), so their agreement is weaker evidence than independent provenance.

5. Methodology

5.1 Unit of analysis and exposure

The unit is the team-match. The outcome is the number of VAR-nullified goals a team scored in that match. Counts are compared against several exposures because no single one is adequate: matches played, shots, shots on target, xG, offside flags (the offside-goal opportunity), corners and *goal events* (goals plus nullified goals). Using several exposures protects against conclusions that depend on the choice of denominator.

5.2 Statistical tests

Test	Question	Method
A. Rate ratios	Is Real Madrid's rate higher?	Exact conditional Poisson test (binomial given the total) for the ratio of two rates, with exact CI; run for each exposure and for two scopes (all 20 teams; top-10 of the final table).
B. Poisson regression	How much of the effect do exposure variables absorb?	GLM with log link, Real Madrid indicator plus log(offsides+1), log(shots+1), xG, home and season fixed effects; standard errors clustered by match; Pearson dispersion reported (1.03–1.11, so Poisson is adequate). Extended with PPDA (own and opponent, att and def, logged) and with fouls, attacking-half fouls, yellow and red cards, always on a common complete-case sample so Real Madrid coefficients are comparable across specifications.
C. Leave-Real-Madrid-out forecast	How many would Real Madrid have if it behaved like the others?	Fit models on the other teams only, predict Real Madrid's 312 matches, simulate the total with gamma-Poisson draws (20,000) when dispersion > 1, report the 95% interval and the probability of observing at least 50.
D. Within-match symmetry	In the same matches (same referee, same VAR room, same pitch and day), is Real Madrid's share of the nullified goals what its attack share predicts?	Conditional on the match total, simulate the Real Madrid count as binomial with probability equal to Real Madrid's share of offsides, shots, xG or goal opportunities in that match (20,000 draws). A bias against Real Madrid would reject the offside and goal-opportunity nulls.
E. Favouritism	Are Real Madrid's opponents spared?	Compare nullified goals for opponents in matches against Real Madrid with their own season rates (per match, per offside, per shot); Poisson test for fewer than expected.

Test	Question	Method
F. Referee heterogeneity	Is the effect concentrated in particular referees?	Chi-square test of homogeneity of Real Madrid's rate across the 15 referees with at least 8 Real Madrid matches; comparison with the same referee's rate for other teams.
G. VAR vs on-field stage	Is the Real Madrid excess specific to VAR?	Rate tests and Poisson models on on-field disallowed goals, and a stacked model with a Real Madrid × stage interaction.
H. Correlations	Do attack volume and offsides track nullified goals across top-10 team-seasons?	Pearson, Spearman and Kendall coefficients with 2,000-sample bootstrap CIs; with and without Real Madrid; at team-season and team-aggregate level.
I. Title scenarios	Would the nullified goals have changed any title?	Rebuild each season from all 380 results, add back goals, recompute the whole table and re-rank with the La Liga tie-break; break-even analysis (dynamic programming over the matches) and a Monte-Carlo in which each nullification is wrong with probability p.

5.3 Counterfactual scenarios

- **A – Real Madrid only (VAR):** add back all 50 Real Madrid VAR-nullified goals. This is an upper bound: it assumes every nullification was wrong.
- **B – Everyone (VAR):** add back every team's VAR-nullified goals, i.e. a league in which VAR never nullifies a goal.
- **C – A plus on-field decisions:** an extreme sensitivity that also treats goals ruled out on the pitch as valid.

When goals are added to a match, the opponent's points change accordingly, so the rival's total in the same match is recomputed.

5.4 Software

Python 3.12; pandas, numpy, scipy (exact tests, bootstrap), statsmodels (GLM with cluster-robust covariance), matplotlib/seaborn, xlsxwriter. All steps are scripted and seeded (seed 2026).

6. Results

6.1 Ranking and composition

Real Madrid has 50 VAR-nullified goals in 312 league matches, ahead of Villarreal (30), Barcelona (27), Atlético (26), Sevilla (25) and Celta (23) (Figure 3). Offside is the dominant reason league-wide and for Real Madrid (36 of 50; Table 5). Real Madrid's count is spread across eight seasons, peaking in 2022-23 (11) and 2024-25 (9) (Figure 4).

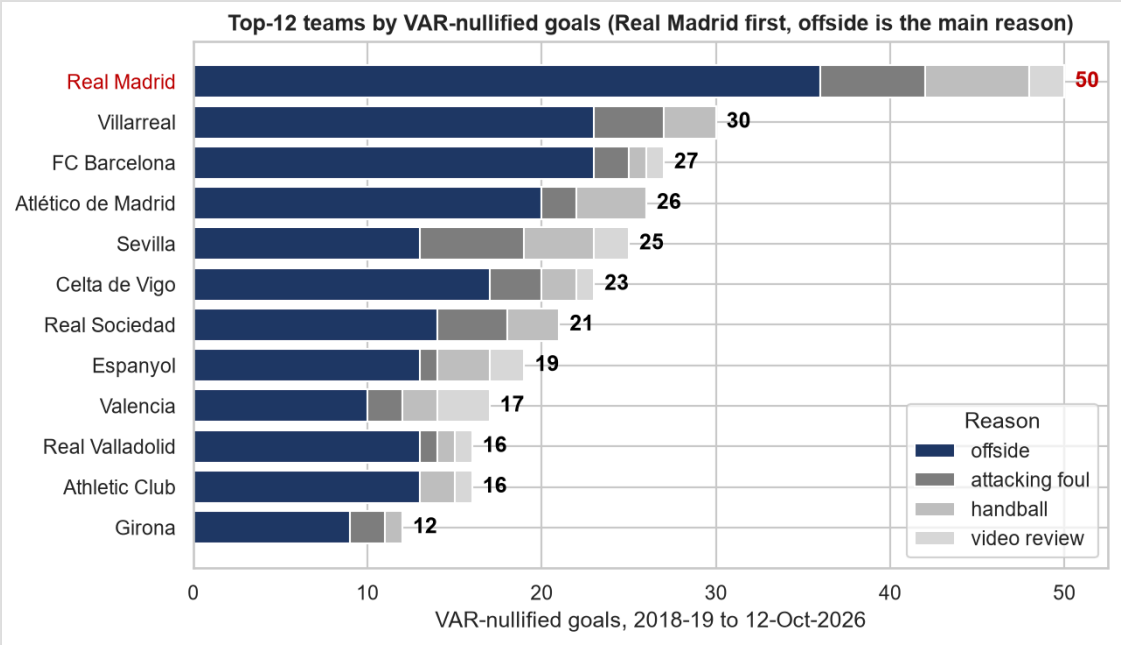


Figure 3. VAR-nullified goals by team and reason, 2018-19 to 12-Oct-2026 (top 12).

Table 5. VAR-confirmed nullified goals by reason, top 8 teams.

Team	Offside	Attacking foul	Handball	Video review (unspec.)	Total
Real Madrid	36	6	6	2	50
Villarreal	23	4	3	0	30
FC Barcelona	23	2	1	1	27
Atlético de Madrid	20	2	4	0	26
Sevilla	13	6	4	2	25
Celta de Vigo	17	3	2	1	23
Real Sociedad	14	4	3	0	21
Espanyol	13	1	3	2	19

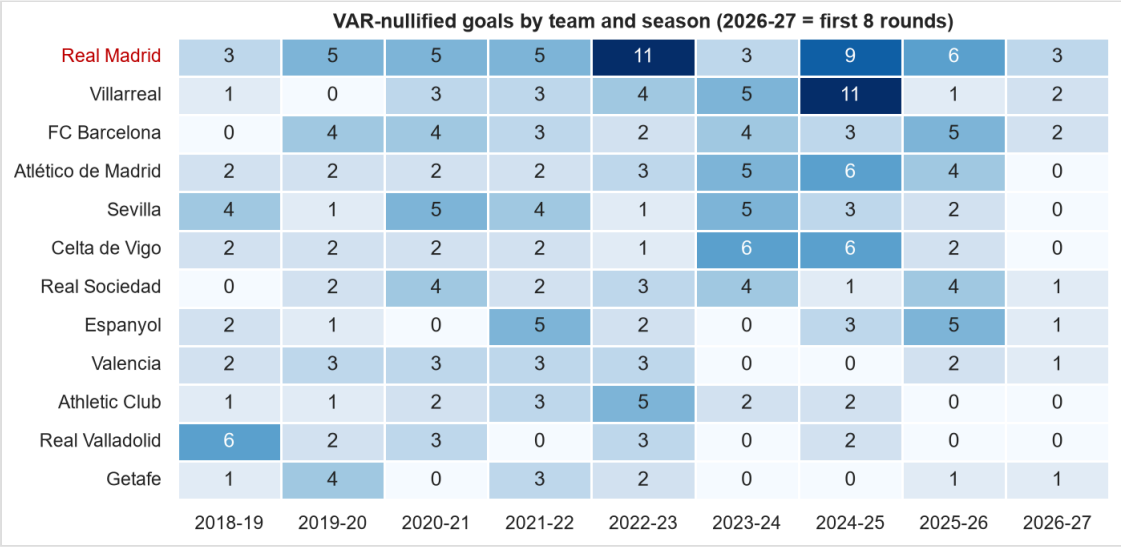


Figure 4. VAR-nullified goals by team and season (2026-27 covers the first 8 rounds).

6.2 The excess is statistically unambiguous

Per match, Real Madrid's rate is 0.160 versus 0.058 for the other teams, a ratio of 2.76 (95% CI 2.01–3.72). The ratio shrinks when the rate is normalised by attacking exposure but is above 1.7 for every exposure we tried (Table 6; top-10 scope: 1.7–2.4). Because Real Madrid was singled out as the most extreme of 31 clubs, we checked that selection does not explain it: even a Bonferroni threshold across 31 teams ($p < 0.0016$) is met by a wide margin (exact one-sided $p = 2.3e-09$), and Real Madrid sits far above the 99.8% limit of a funnel plot of all teams (Figure 5). It is also not an artefact of a few multi-goal matches: 14.1% of Real Madrid's matches contain at least one nullified goal against 5.5% for other teams (odds ratio 2.8; the largest single-match count is 3).

Table 6. Real Madrid versus all other teams: rate ratios by exposure (exact conditional Poisson test; all $p < 0.002$).

Exposure	RM events	RM rate	Others events	Others rate	Rate ratio	CI low	CI high
per match played	50	0.160	344	0.058	2.76	2.01	3.72
per 100 shots	50	0.990	344	0.498	1.99	1.45	2.68
per 100 shots on target	50	2.610	344	1.452	1.80	1.31	2.42
per 10 xG	50	0.785	344	0.437	1.79	1.31	2.42
per 100 offsides	50	6.803	344	2.815	2.42	1.76	3.26
per 100 corners	50	2.729	344	1.252	2.18	1.59	2.94
per 100 goal events	50	7.508	344	4.427	1.70	1.23	2.29

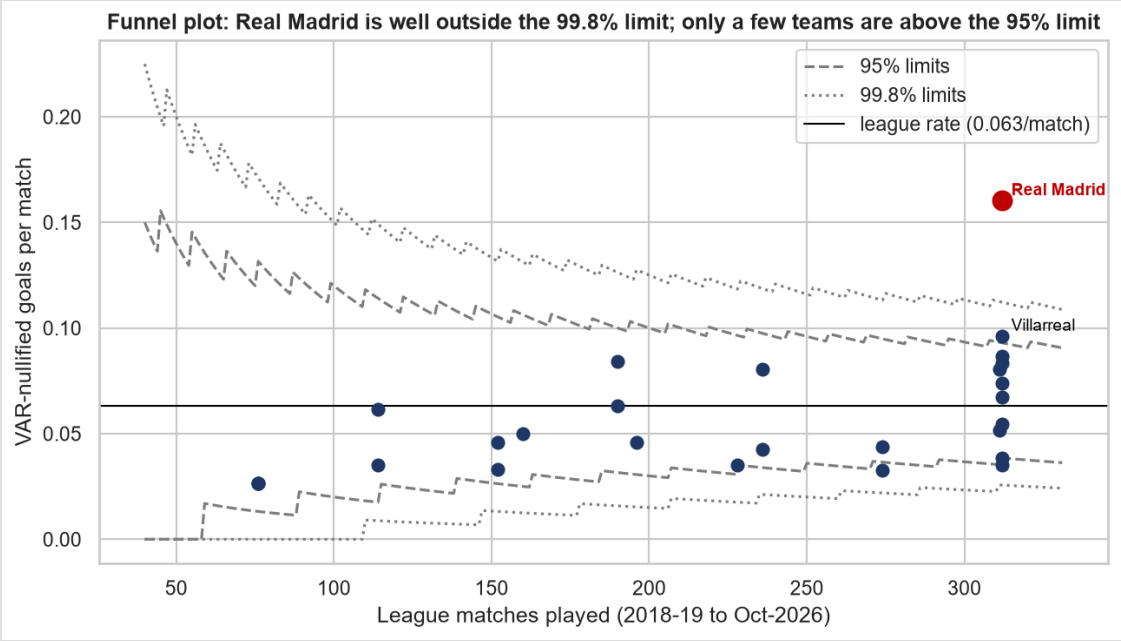


Figure 5. Funnel plot: nullified goals per match against matches played, with 95% and 99.8% Poisson limits around the league rate. Real Madrid is far outside the 99.8% limit; only a few clubs exceed the 95% limit.

6.3 How much does attacking volume explain?

Correlations (top-10 team-seasons). Across the top-10 team-seasons, nullified goals correlate moderately with chances created (shots $r = 0.48$, 95% CI 0.30–0.63; xG $r = 0.49$) and only weakly with offside flags ($r = 0.21$, $p = 0.06$). Excluding Real Madrid, the figures are 0.30 (shots), 0.38 (xG) and 0.20 (offsides): the association does not depend on Real Madrid alone, but Real Madrid sits well above the fitted line (Figure 6). Offside flags per match are close to the top-10 average (2.36 vs 2.14), so offside *volume* by itself does not single Real Madrid out.

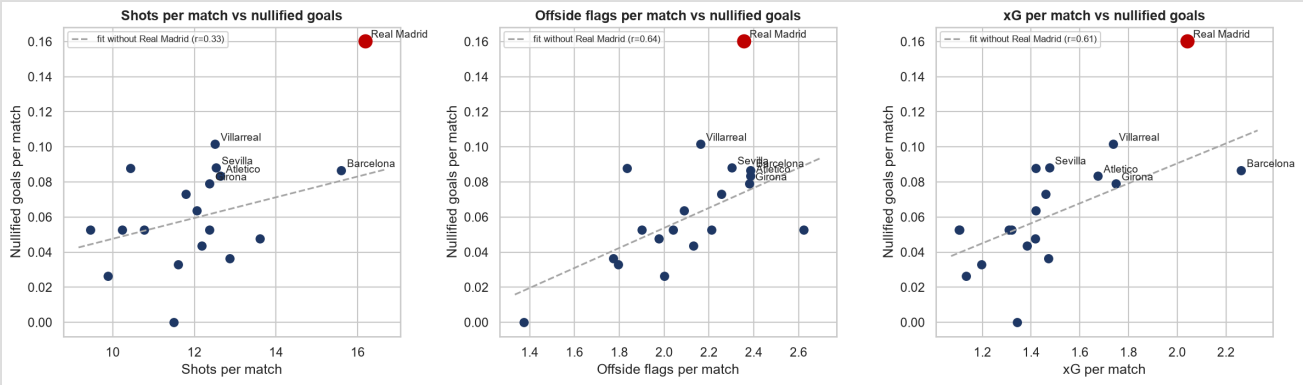


Figure 6. Nullified goals per match against shots, offside flags and xG per match for top-10 teams (aggregates 2018-2026). Dashed line: fit without Real Madrid.

Regression. In Poisson models Real Madrid's incident-rate ratio (IRR) falls from 2.74 with no controls to 2.01 after controlling for offsides, shots, xG, home and season (95% CI 1.45–2.79; top-10 scope 1.84, Table 7, Figure 7). Offside flags are the single strongest covariate (about 2.4 times more nullified goals per log-unit of offsides), but neither offsides nor shots nor xG absorbs the Real Madrid effect.

Table 7. Poisson regression: Real Madrid incident-rate ratio under progressively richer controls (SE clustered by match).

Model (all 20 teams, 6,160 team-matches)	RM IRR	CI low	CI high	p	Dispersion
raw: RM indicator only	2.74	1.98	3.79	<0.001	1.07
+ offsides	2.52	1.83	3.48	<0.001	1.03
+ shots	2.20	1.57	3.07	<0.001	1.06
+ shots on target	2.28	1.64	3.16	<0.001	1.06
+ xG	2.24	1.61	3.11	<0.001	1.06
+ offsides + shots	2.03	1.46	2.82	<0.001	1.03
+ offsides + shots + xG + home + season FE	2.01	1.45	2.79	<0.001	1.03

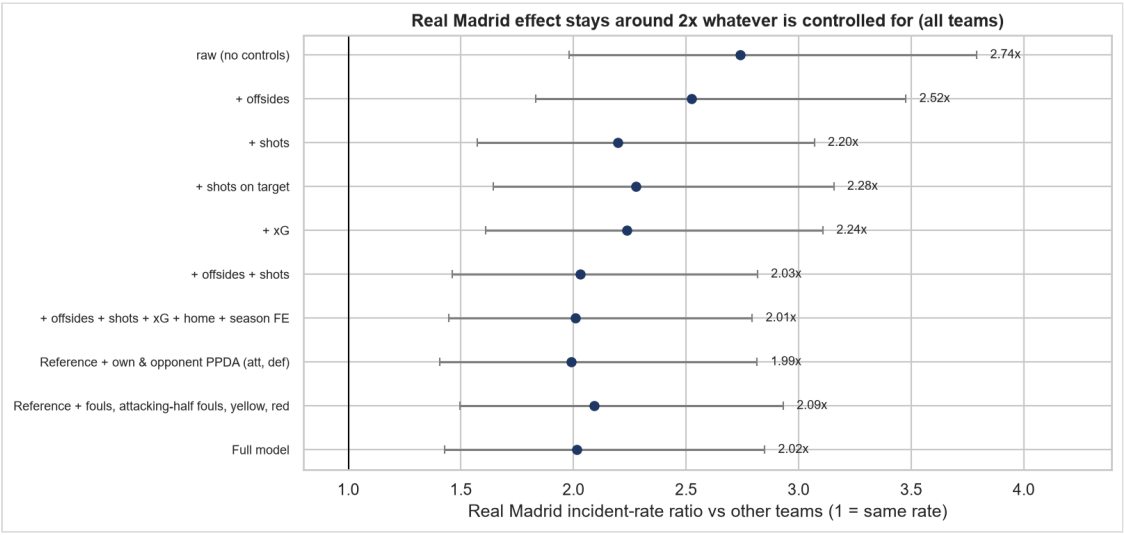


Figure 7. Real Madrid incident-rate ratio (95% CI) across specifications, all teams. The effect remains near 2 whatever is controlled for.

Forecast. A model trained *only on the other 30 clubs* (offsides, shots, xG and home) predicts about 25 nullified goals for Real Madrid's 312 matches (95% interval 16–35), against 50 observed (Table 8, Figure 8). Attack volume therefore explains roughly half of the count; the remaining 25 goals are about twice what that model expects.

Table 8. Leave-Real-Madrid-out forecast of Real Madrid's total (20,000 gamma-Poisson simulations; all p(obs or more) < 0.001).

Model trained on other teams	Expected	2.5%	97.5%	Excess (obs 50)	Obs / expected
matches only (league average rate)	18.2	10	27	31.8	2.74
offsides	19.8	12	29	30.2	2.53
shots	22.2	13	32	27.8	2.25
xG	22.0	13	32	28.0	2.27
offsides + shots + xG	25.0	16	35	25.0	2.00

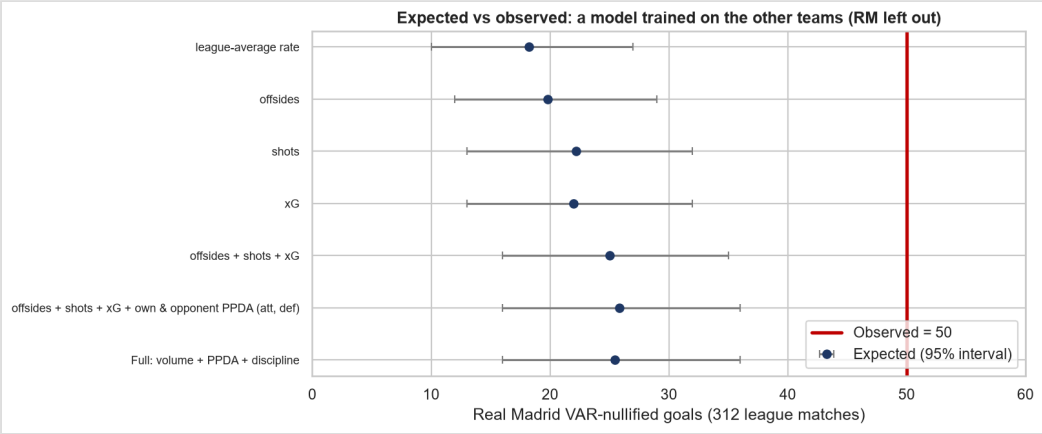


Figure 8. Expected versus observed Real Madrid nullified goals under models trained on the other teams (95% intervals). The observed 50 (red line) is outside every interval.

6.4 Pressing, fouls and cards

Real Madrid's profile differs from the other top-10 teams in several ways (Figure 9): more shots (1.31×), shots on target (1.40×), xG (1.32×), goals (1.33×), possession (1.14×) and corners (1.19×); about 1.30× more passes in the opponent's pressing zone; opponents press it much less (opponent PPDA 14.8 versus 11.1); and it commits fewer fouls (10.4 vs 12.9 per match; 5.7 vs 7.0 in the attacking half) and receives fewer yellow cards (1.88 vs 2.35); its own pressing intensity (PPDA 10.5 vs 10.0) and red cards (0.10 vs 0.11) are typical.

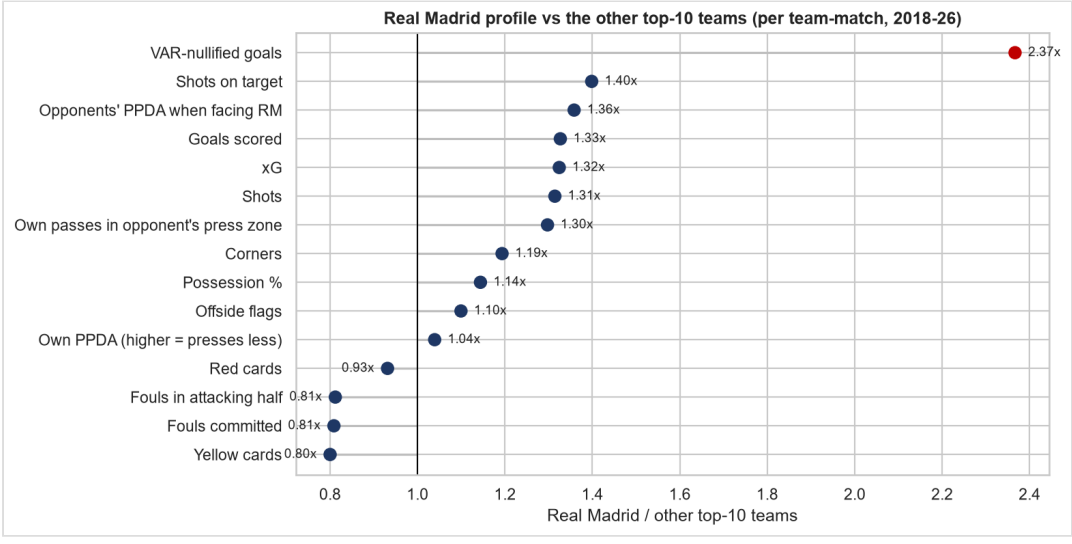


Figure 9. Real Madrid relative to the other top-10 teams, per team-match, 2018-2026 (ratio of means; 1 = same). The nullified-goal ratio (2.4×) exceeds every other profile ratio.

Adding own and opponent PPDA (attempts and defensive actions, logged) moves the Real Madrid IRR from 2.06 to 1.99 (6,006 team-matches; Table 9). Adding total fouls, attacking-half fouls, yellow and red cards moves it from 2.06 to 2.02 when PPDA is also included (5,980 team-matches). The model trained on other teams with all of these covariates expects ~25 goals (95% 16–36). Individual PPDA and discipline coefficients are imprecise (PPDA components are collinear by construction and most intervals include 1); the informative finding is the stability of the Real Madrid coefficient. The negative team-level correlation between fouls and nullified goals (r = −0.40 across top-10 team-seasons) is best read as possession-dominant teams fouling less and attacking more, not as discipline causing offsides.

Table 9. Effect of adding pressing context (own and opponent PPDA).

PPDA models (all teams)	n	RM IRR	CI low	CI high
Reference (offsides + shots + xG + home + season FE), PPDA-complete sample	6,006	2.06	1.48	2.86
+ own PPDA att and def	6,006	2.09	1.49	2.94
+ opponent PPDA att and def	6,006	1.91	1.36	2.68
+ own and opponent PPDA att and def	6,006	1.99	1.41	2.82

Table 10. Effect of adding fouls, fouls in the attacking half, yellow and red cards (and PPDA).

Discipline models (all teams)	n	RM IRR	CI low	CI high
Reference (offsides + shots + xG + home + season FE)	5,980	2.06	1.48	2.86
+ total fouls committed	5,980	2.06	1.47	2.89
+ fouls committed in attacking half	5,980	2.05	1.47	2.87
+ yellow cards + red cards	5,980	2.12	1.53	2.94
+ total fouls + yellow + red	5,980	2.09	1.49	2.93
+ attacking-half fouls + yellow + red	5,980	2.11	1.51	2.94
+ total fouls + attacking-half fouls + yellow + red	5,980	2.09	1.49	2.93
+ fouls (total, attacking half) + yellow + red + own & opponent PPDA	5,980	2.02	1.43	2.85

6.5 Does the residual carry the fingerprints of targeting or favouritism?

Within-match symmetry (test D). If officials leaned against Real Madrid, then in Real Madrid matches (same referee, same VAR room) Real Madrid would lose *more* goals than its offside or goal-opportunity share predicts. It does not: Real Madrid has 50 of the 63 nullified goals in its matches (79%); its offside share predicts 46 (95% 41–51; p = 0.11) and its share of goal opportunities 46 (p = 0.14). Under shots and xG shares Real Madrid is above expectation (p = 0.002 and 0.014), which is the same signal as in the regressions: Real Madrid produces more offside goals than its shots imply, but not more than its offsides and goal opportunities imply.

Table 11. Within-match test: observed Real Madrid share of nullified goals in its own matches versus attack-share null models (20,000 simulations).

RM share of nullified goals predicted by	RM obs.	Opponents obs.	RM expected	2.5%	97.5%	p (RM >= obs.)
Share of offsides	50	13	46.1	41	51	0.110
Share of shots	50	13	39.1	32	46	0.002
Share of xG	50	13	42.3	35	49	0.014
Share of goal opportunities (goals + nullified)	50	13	46.3	41	52	0.135

Favouritism (test E). Opponents have 13 goals nullified in matches against Real Madrid versus 18.7 expected from their own per-match rates (ratio 0.70, p = 0.11); after allowing for their offside count the ratio is 1.14. There is no statistical sign that opponents are systematically protected.

Table 12. Are Real Madrid's opponents spared? Opponents' nullified goals in matches against Real Madrid versus expectation from their own season rates.

Baseline for opponents	Observed	Expected	Obs / exp	p (fewer than expected)
their own per-match rate that season	13	18.7	0.70	0.112
their own per-offside rate that season	13	11.4	1.14	0.746
their own per-shot rate that season	13	17.1	0.76	0.193

Referees (test F). Across the 15 referees with at least eight Real Madrid matches, Real Madrid's rate is homogeneous (χ^2 = 11.9, 14 d.f., p = 0.62), and it exceeds the same referee's rate for other teams for 12 of the 15, a league-wide pattern rather than a one-referee story (Figure 10, right). Per-referee samples are small (0–8 events), so the homogeneity test, not individual ratios, is the relevant evidence.

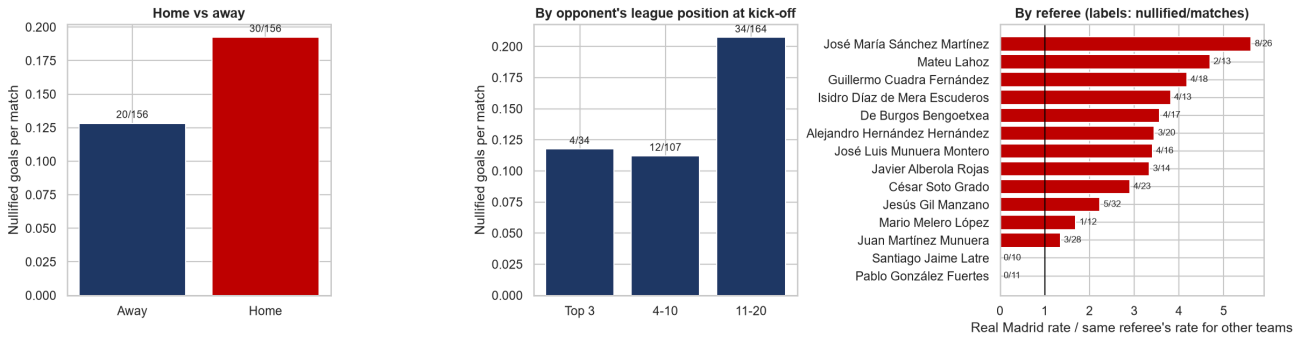


Figure 10. Real Madrid nullified goals per match by venue, by the opponent's league position at kick-off, and by referee (ratio against the same referee's rate for other teams; labels show events/matches).

Net points ledger. Counting, for every team, the points lost to its own nullified goals and the points kept because an opponent's goal was nullified, Real Madrid lost 21 points and kept 12: a net of -9, similar to Villarreal (–9) and Barcelona (–7), while other teams such as Athletic Club (+14) benefited on net (Table 13, Figure 11). This is a measure of exposure to nullified goals, not a causal estimate of bias.

Table 13. Net points from nullified goals, 2018-2026 (counterfactual per event, all else equal).

Team	Own goals nullified	Points lost	Opponents' goals nullified vs team	Points kept	Net points
Athletic Club	16	9	27	23	+14
Sevilla	25	14	25	22	+8
Real Sociedad	21	12	21	16	+4
Atlético de Madrid	26	11	9	10	-1
FC Barcelona	27	13	14	6	-7
Real Madrid	50	21	13	12	-9
Villarreal	30	19	14	10	-9

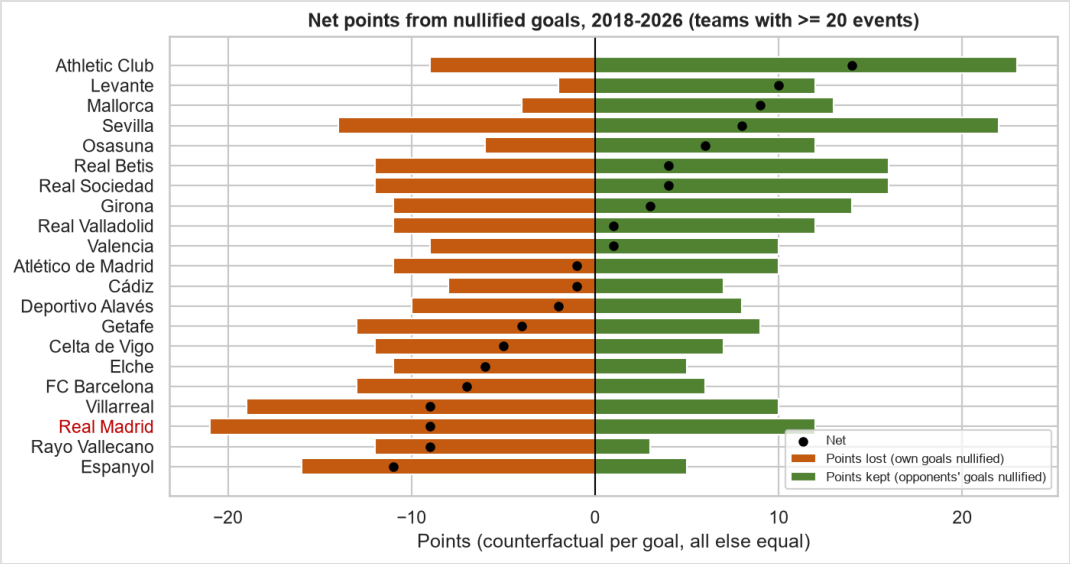


Figure 11. Points lost to a team's own nullified goals (orange), points kept because an opponent's goal was nullified (green) and net (dot); teams with at least 20 events.

6.6 Is the excess specific to VAR?

Goals ruled out by on-field officials without any VAR review show the same direction: Real Madrid has 29 of the 298 (rate ratio 2.0 raw, 1.4 after controls, p = 0.07), alongside Barcelona (28), Atlético (27) and Villarreal (26) (Figure 12). Combining both stages gives 79 Real Madrid events (2.4×). In a stacked model, the extra Real Madrid effect at the VAR stage compared with the on-field stage is 1.36× (95% CI 0.82–2.23, p = 0.23). The point estimate is above 1, so a modest VAR-specific effect cannot be excluded with 50 VAR events, but the data show no evidence of one: the elevated rate already exists before VAR is involved.

Table 14. Real Madrid versus other teams: on-field disallowed goals (no VAR), VAR-nullified goals, and both combined (per match).

Category	RM events	RM / match	Others events	Others / match	Rate ratio	CI low	CI high
On-field only (no VAR)	29	0.093	269	0.045	2.05	1.34	3.01
VAR-nullified only	50	0.160	344	0.058	2.76	2.01	3.72
Combined (on-field + VAR)	79	0.253	613	0.103	2.45	1.91	3.10

Table 15. VAR stage versus on-field stage: Real Madrid x stage interaction (all teams; stacked counts, SE clustered by match).

Model	Term	IRR	CI low	CI high	p
Raw	Real Madrid effect on on-field disallowed goals	2.02	1.38	2.95	<0.001
Raw	Extra Real Madrid effect at the VAR stage vs on-field	1.36	0.82	2.23	0.231
Controlled (offsides, shots, xG, home, season FE)	Real Madrid effect on on-field disallowed goals	1.46	1.00	2.12	0.049
Controlled (offsides, shots, xG, home, season FE)	Extra Real Madrid effect at the VAR stage vs on-field	1.36	0.82	2.23	0.231

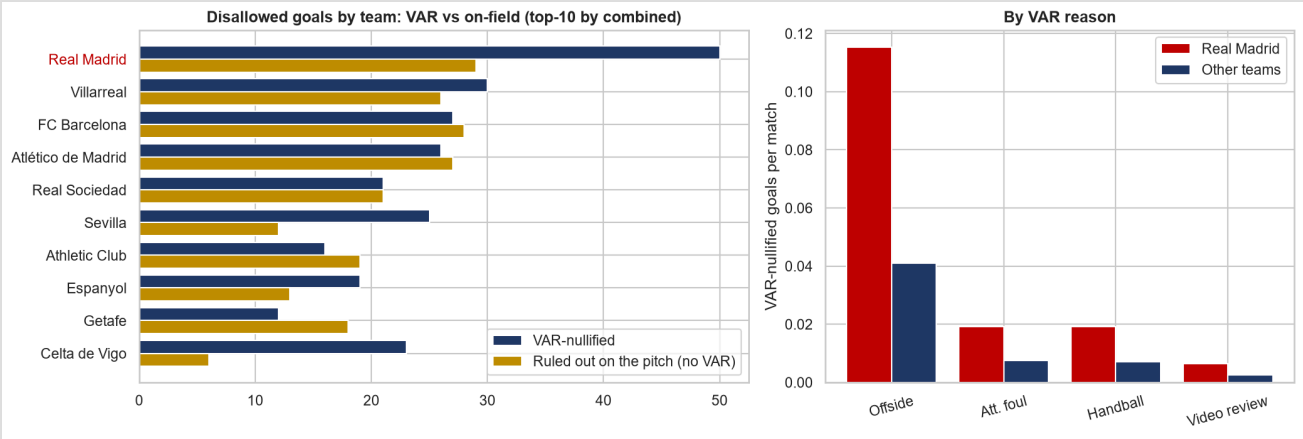


Figure 12. Left: disallowed goals by team, VAR-nullified (navy) versus ruled out on the pitch (gold). Right: VAR-nullified goals per match by reason, Real Madrid versus other teams.

By reason, the excess is dominated by offside (36 events, 2.8× the rate of others; 95% CI 1.9–4.0); attacking foul (6, 2.6×) and handball (6, 2.7×) have too few events for their intervals to exclude 1.

Table 16. VAR-nullified goals by reason: Real Madrid versus all other teams.

VAR reason	RM events	RM / match	Others / match	Rate ratio	CI low	CI high
Offside	36	0.115	0.041	2.81	1.92	4.00
Attacking foul	6	0.019	0.007	2.59	0.90	6.10
Handball	6	0.019	0.007	2.71	0.94	6.41
Video review (unspecified)	2	0.006	0.003	2.53	0.28	10.89
All reasons	50	0.160	0.058	2.76	2.01	3.72

6.7 Descriptive profile of the 50 goals

Players. The goals involve 23 different scorers; 11 of them account for 75% of the incidents (76%; Figure 13). Karim Benzema and Kylian Mbappé lead with 8 each (32% together), followed by Marco Asensio and Vinícius Júnior with 4 each. Concentration is typical for a team of this size: the top-three share is 40% against a median of 38% for the other teams with at least 15 nullified goals (Figure 14). Normalising by offside flags, the conversion of a flag into a nullified goal is highest for Asensio and Valverde (small denominators) and about 7.7 per 100 flags for Mbappé, 4.5 for Benzema and 1.9 for Vinícius (Figure 15).

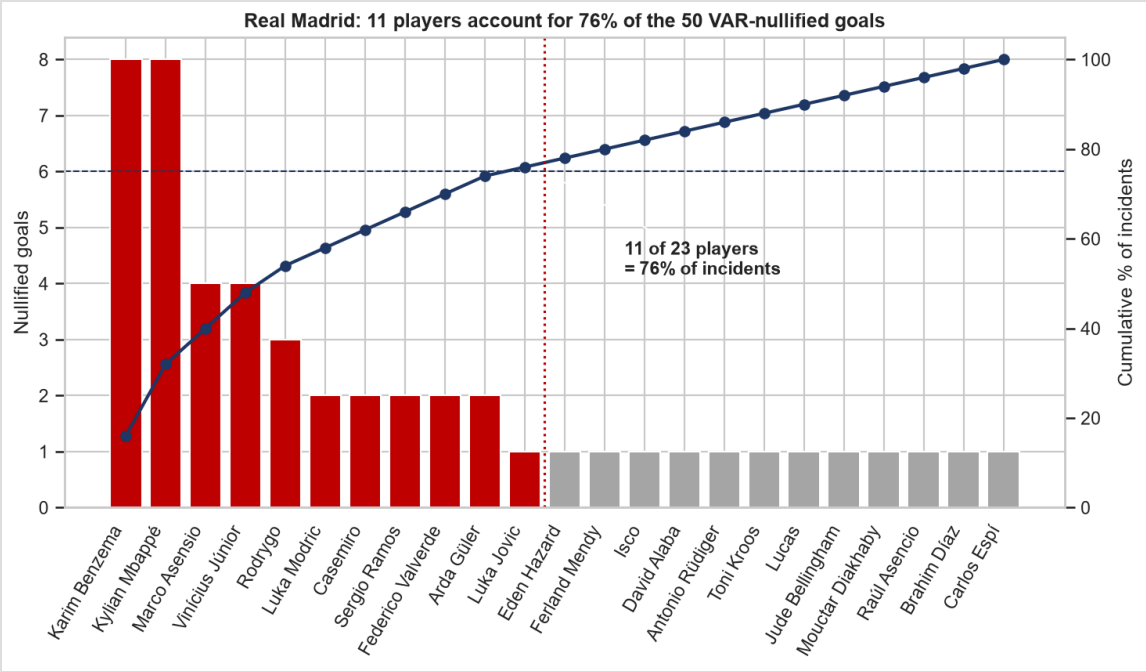


Figure 13. Real Madrid VAR-nullified goals by player (bars) and cumulative share (line). Eleven of 23 players (red) account for 76% of the incidents.

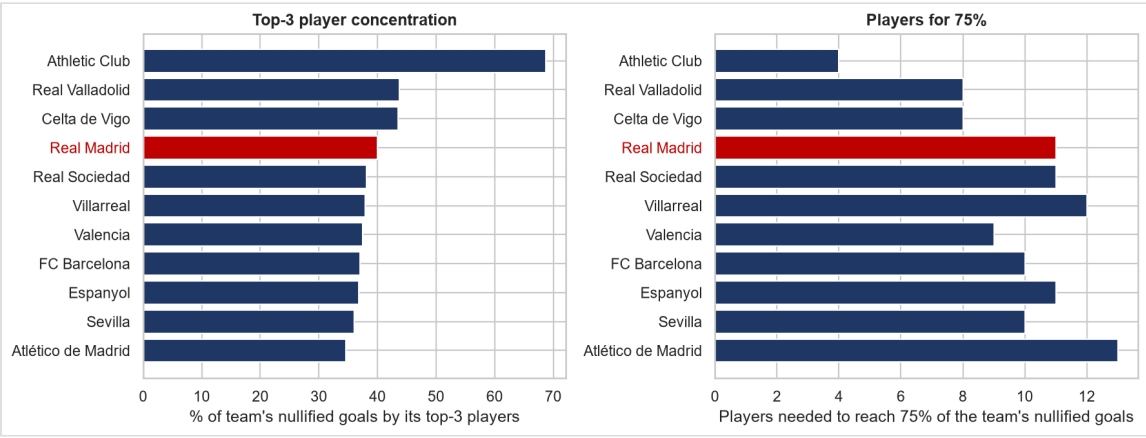


Figure 14. Concentration of nullified goals by player across teams with 15 or more events: top-three share and number of players needed to reach 75%.

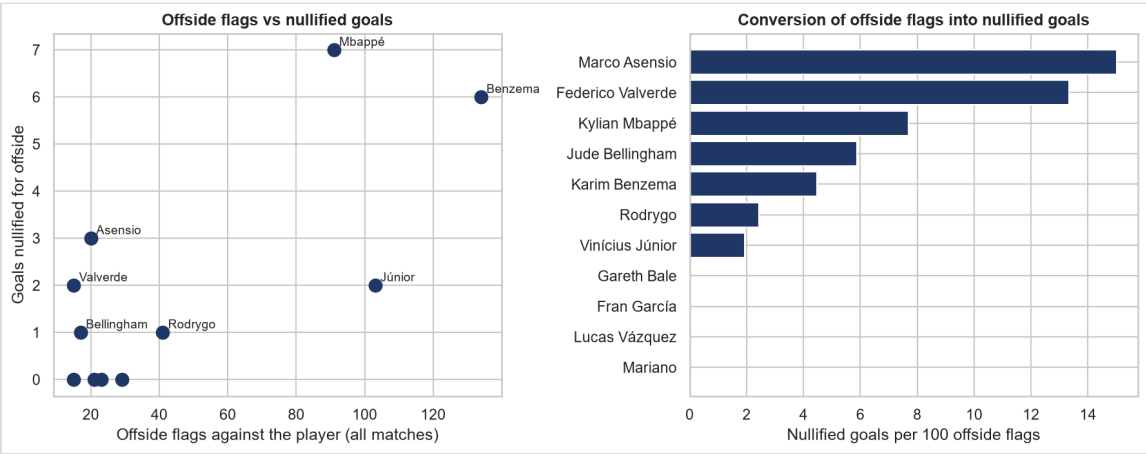


Figure 15. Offside flags against Real Madrid players (ESPN events) versus goals nullified for offside; right: nullified goals per 100 offside flags.

Timing and game state. Real Madrid's nullified goals are spread across the match (54% in the second half versus 53% for other teams); 22% fall in the first 15 minutes versus 13% for others, but with 11 events this is not significant (binomial p = 0.09). 23 of the 50 goals would have been openers and 6 equalisers, 19 would have extended a lead and 2 reduced a deficit (Figure 16).

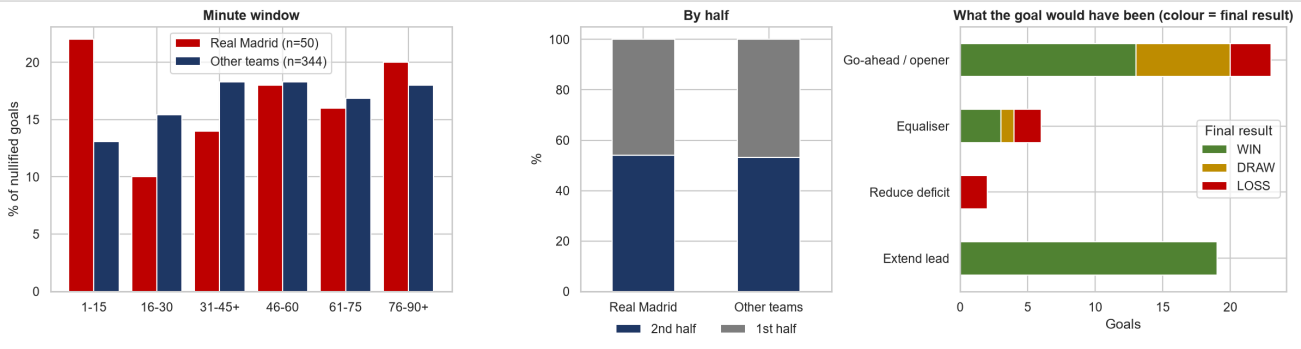


Figure 16. Minute window, half and what the goal would have meant (colour: final result), Real Madrid versus other teams.

Venue and opponent. Thirty of the 50 goals came at home against 20 away (0.192 vs 0.128 per match), but the difference is not significant (p = 0.20) and is present league-wide (other teams: 0.068 home vs 0.048 away). Against opponents outside the top 10 at kick-off Real Madrid's rate is 0.207 versus 0.113 against the top 10 (p = 0.05, borderline), and the same gradient exists for all teams (0.075 versus 0.051); deeper-defending sides concede more goals and more offside goals. None of these cuts is specific to Real Madrid.

Time. The count is irregular across seasons (Figure 17) and strongly clustered in two seasons (11 in 2022-23, 9 in 2024-25), which is why the champion-versus-not comparison below is imprecise.

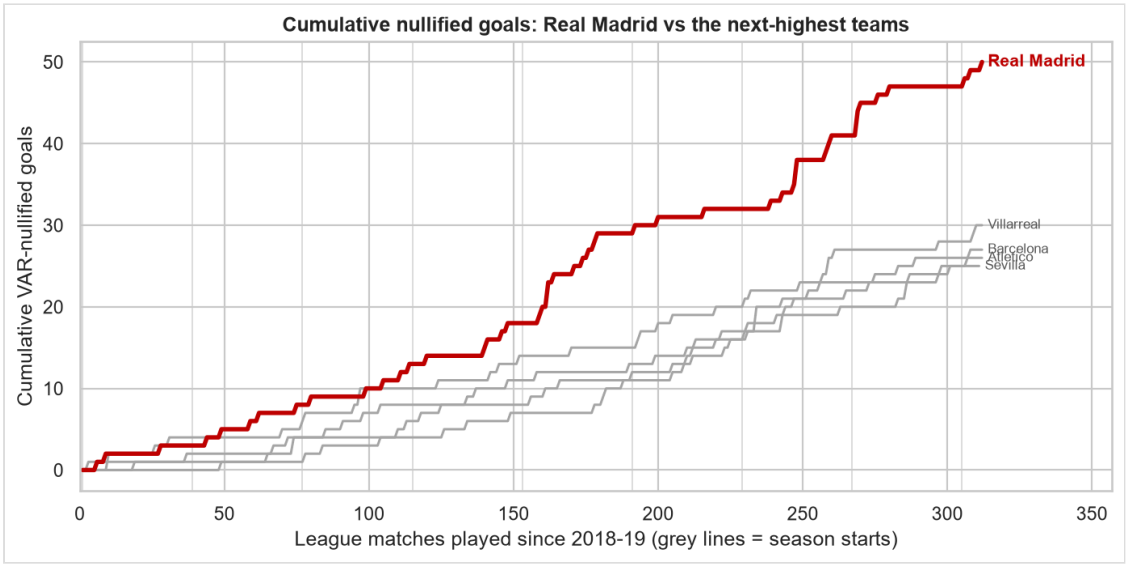


Figure 17. Cumulative VAR-nullified goals by league match since 2018-19: Real Madrid and the four next-highest teams.

6.8 Did it cost titles? Games affected, champion versus not, and scenarios

Matches directly affected. The 50 goals fall in 44 matches (14.1% of Real Madrid's league matches): 29 were wins (35 goals), 8 draws and 7 losses. Of the 15 matches that ended in a draw or loss, in 13 the result would have changed if every nullified goal had counted (8 draws would have been wins, 5 of the 7 losses draws), worth 21 points over 8.2 seasons (about 2.6 per season; Figure 18). Two losses (including the 0–4 Clásico) would not have changed.

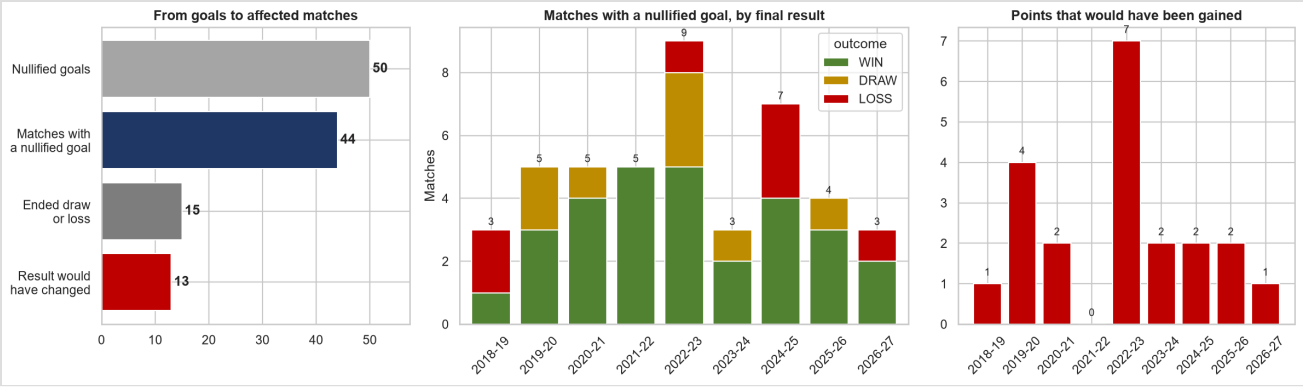


Figure 18. Left: from nullified goals to directly affected matches. Middle: matches with a nullified goal by final result. Right: points that would have been gained if all nullified goals had counted.

Champion versus not champion. Real Madrid won three titles in the period (2019-20, 2021-22, 2023-24). In those seasons it had 13 nullified goals (4.3 per 38 matches; 26% of the total) against 34 in the five non-title seasons (6.8 per 38; 68%); the remaining 3 fall in the partial 2026-27 season. The rate is lower in title seasons but the difference is not significant (rate ratio 0.64, exact two-sided p = 0.18; Figure 19).

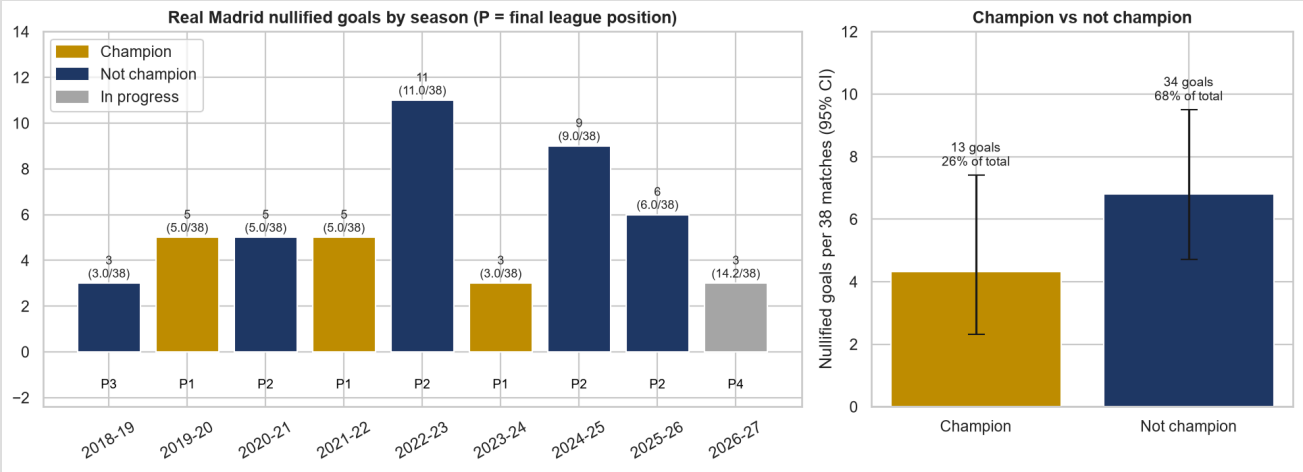


Figure 19. Left: nullified goals per season (P = final position). Right: rate per 38 matches in champion versus non-champion seasons (95% CI).

Title scenarios. Every season was rebuilt from all 380 results and re-ranked with the La Liga tie-break; the rebuilt actual tables reproduce the real champions in all eight full seasons. Counting *all* Real Madrid nullified goals as valid (scenario A, an upper bound) changes exactly one title: **2020-21**, where Real Madrid (84) would finish level on 86 points with Atlético and ahead on head-to-head (4 points to 1); that requires one wrong call, a draw turned into a win. In 2022-23 Real Madrid would still finish one point behind Barcelona (85 vs 86), in 2024-25 two behind (86 vs 88), in 2025-26 six behind (88 vs 94) and in 2018-19 eighteen behind (Table 17, Figures 20–21). In the neutral scenario B (every team's nullified goals count), Real Madrid gains 2022-23 but loses 2023-24 (87 vs 88): the effects roughly offset. Only the extreme scenario C, which also reverses every on-field decision, would give Real Madrid six titles instead of three.

Table 17. Title scenarios (A: all Real Madrid nullified goals counted; B: every team's; C: A plus on-field decisions).

Season	Champion (actual)	RM pts (actual)	RM pts, A	RM pos, A	Champion, A	Champion, B	Champion, C
2018-19	Barcelona	68	69	3	Barcelona	Barcelona	Barcelona
2019-20	Real Madrid	87	91	1	Real Madrid	Real Madrid	Real Madrid
2020-21	Atlético	84	86	1	Real Madrid	Real Madrid	Real Madrid
2021-22	Real Madrid	86	86	1	Real Madrid	Real Madrid	Real Madrid
2022-23	Barcelona	78	85	2	Barcelona	Real Madrid	Real Madrid
2023-24	Real Madrid	95	97	1	Real Madrid	Barcelona	Real Madrid
2024-25	Barcelona	84	86	2	Barcelona	Barcelona	Real Madrid
2025-26	Barcelona	86	88	2	Barcelona	Barcelona	Barcelona

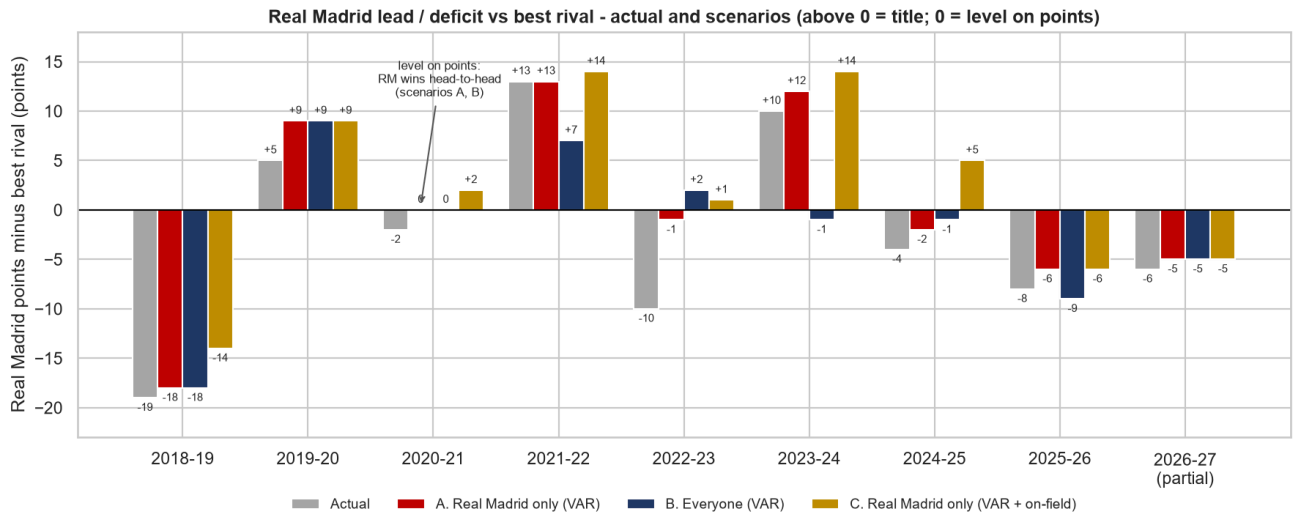


Figure 20. Real Madrid points minus the best rival, actual and under scenarios A, B and C (above 0 = title; 0 = level on points, settled by head-to-head).

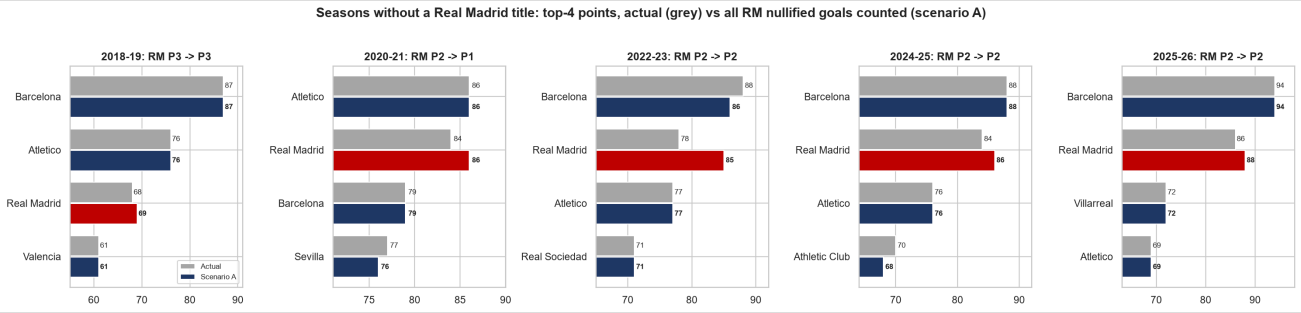


Figure 21. Seasons without a Real Madrid title: top-four points, actual (grey) versus scenario A. Only 2020-21 changes the champion.

How many calls would have to be wrong? A break-even analysis (maximising the points Real Madrid and its rival could swing by flipping any subset of nullified goals) shows that in four of five non-title seasons *no* feasible set of wrong calls would be enough to reach the champion's points (Table 18, “not reachable” where Real Madrid's own goals cannot close the gap). In 2020-21 a single wrong call suffices. A Monte-Carlo, in which every nullification is wrong independently with probability p, gives the probability that Real Madrid would have been champion (Table 19, Figure 22): 1.9% at p = 1.7% (the 98.3% accuracy of Spitz et al.), 5.1% at 5%, 9.8% at 10% and 25% at 25%. In every other season it is 0%.

Table 18. Break-even analysis for seasons without a Real Madrid title (RM only; head-to-head decides ties on points).

Season	Champion	Gap (pts)	RM nullified goals	Max points recoverable	Min wrong calls to level	RM wins head-to-head	H2H pts (RM-champion)
2018-19	Barcelona	19	3	1	not reachable	False	0-6
2020-21	Atlético	2	5	2	1	True	4-1
2022-23	Barcelona	10	11	9	not reachable	True	3-3
2024-25	Barcelona	4	9	2	not reachable	False	0-6
2025-26	Barcelona	8	6	2	not reachable	False	3-3

Table 19. Monte-Carlo: probability (%) that Real Madrid would have been champion if each nullified goal were wrong with probability p (20,000 simulations).

Season	p = 1.7%	p = 5.0%	p = 10.0%	p = 25.0%	p = 50.0%
2018-19	0.0	0.0	0.0	0.0	0.0
2020-21	1.9	5.1	9.8	24.7	49.6
2022-23	0.0	0.0	0.0	0.0	0.0
2024-25	0.0	0.0	0.0	0.0	0.0
2025-26	0.0	0.0	0.0	0.0	0.0

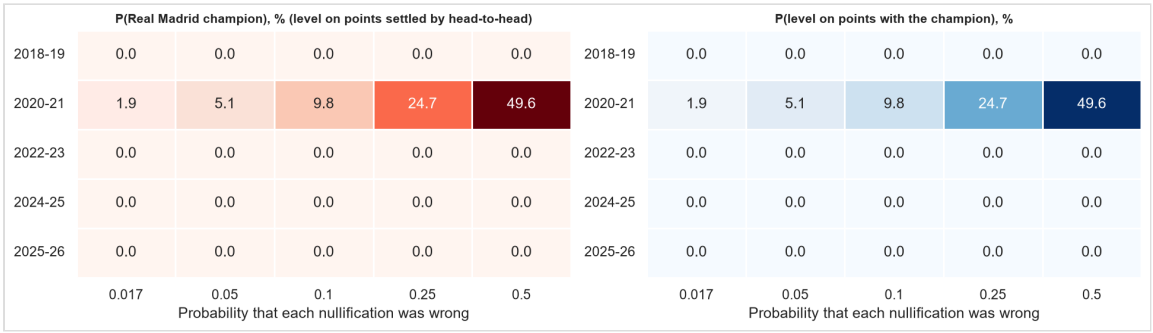


Figure 22. Monte-Carlo probability that Real Madrid would have been champion (left) or level on points (right) as a function of the VAR error rate.

7. VAR accuracy in context

The brief for this study set a prior that VAR is more than 95% accurate and asked for La Liga evidence first, then the Premier League, then FIFA. We could not retrieve and verify a La Liga-specific accuracy publication. The evidence we could verify or locate is: (i) the peer-reviewed multi-league study of Spitz et al. (2021), covering 2,195 matches in 13 countries, with 9,732 VAR checks and 795 reviews, in which the accuracy of the final decision rose from 92.1% (initial referee call) to 98.3% after VAR (abstract verified via Europe PMC, PMID 32794432); (ii) FIFA's reported success rate of 99.3% at the 2018 World Cup, against 95% without VAR (as cited by Wikipedia); and (iii) headline-level reports that the Premier League claims 96% correct decisions (Sky Sports, February 2024) and a 98.9% figure in the IFAB experiment report (BBC, January 2018), and a claim by Spanish refereeing bodies of 98% accuracy on penalties (El País, December 2020). The last three were verified at headline level only. Spanish media have also published rankings of clubs “benefited and harmed” by VAR; their methods could not be retrieved and they were not used.

These accuracy figures matter in two ways. First, at 98.3% accuracy we would expect 0.9 wrong decisions among Real Madrid's 50 nullifications, 2.0 at 96% and 2.5 at 95%, versus an unexplained excess of about 25 goals; even a 20% error rate (more than twice the 7.9% error of unaided referees in Spitz et al.) would

give only 10 wrong decisions. VAR error cannot be what produces the excess. Second, nullified offsides are geometric calls and the least subjective VAR category, and they dominate Real Madrid's count (72%). A bias story would have to operate through *which incidents are flagged or reviewed*, which the tests in Sections 6.5 and 6.6 do not support.

8. Discussion and threats to validity

8.1 Interpretation

The data support a clear, two-part reading. First, Real Madrid is a genuine statistical outlier: the excess is not noise, not a selection artefact, not driven by a few matches and not specific to one data provider. Second, the excess has the properties one would expect of an attacking-profile effect and lacks those that targeting would produce. It is roughly half explained by volume; the remainder is present to a similar degree in decisions taken without VAR; it is symmetric within matches when conditioned on offsides and goal opportunities; it is spread over referees; and opponents are not protected. A plausible mechanism, consistent with the profile data (1.3× shots, 1.3× passes in the opponent's pressing zone, opponents pressing Real Madrid far less) and with the dominance of offside decisions, is that Real Madrid's attackers receive the ball in behind more often and from more advanced positions, creating more goal-then-offside events than shot or offside *counts* capture. We stress that this mechanism is a hypothesis, not a finding.

8.2 Threats to validity

1. **Outcome ascertainment.** Events come from public match feeds, not from an official VAR log. If high-profile clubs' incidents are logged more diligently, counts are differentially inflated for Real Madrid. The cross-check with ESPN (Section 4.3) and the manual video review mitigate this but cannot eliminate it. Early seasons are less well corroborated by ESPN.
2. **Over-control.** Offside flags are themselves produced by the same officiating process whose fairness is in question. Controlling for them (and for the offside-share null in test D) assumes the flags are unbiased; if they were biased the adjusted estimates would absorb part of any bias. The unadjusted 2.8× is the upper bound; the adjusted estimates should be read as conditional on offside flags. That said, the shots-based and xG-based adjustments do not use officiating outcomes and also leave a rate ratio of 1.9–2.2.
3. **Selection through the final table.** "Top-10" is defined by the final table, i.e. conditional on a (partly) post-treatment outcome. It was requested to simplify comparisons; the full-league results are reported alongside and are consistent.
4. **Small counts and multiplicity.** 50 events, nine seasons. Dispersion is modest (1.03–1.11), but p-values from many related tests are descriptive and not corrected for multiplicity. The main outlier result survives Bonferroni by a wide margin; the secondary descriptive cuts (timing, venue, opponent strength) do not and are reported as such.
5. **Dependence.** Standard errors are clustered by match (both teams share a referee), not by team or team-season; team-level clustering would widen the intervals, although the conclusion that the Real Madrid ratio exceeds 1 is robust to that.
6. **Covariate quality.** xG is a model output (Understat), PPDA is a zone-based proxy and foul-zone counts capture about 91% of fouls. PPDA terms are collinear.
7. **Officials.** The recorded referee is the on-field referee, not the VAR official. Heterogeneity by VAR-room official could not be tested.
8. **Counterfactuals.** Points that "would have been gained" assume the match otherwise unfolds identically. They ignore tactical and psychological knock-on effects, and scenario A assumes every

nullification was wrong, which the accuracy literature makes implausible.

9. **Period and protocol effects.** Season fixed effects absorb league-wide changes, but changes in VAR protocol, offside technology and coaching staff (Real Madrid had several head coaches in the period) are not modelled separately.
10. **Causal identification.** The design is observational; absence of evidence for bias is not evidence of absence, particularly with 50 events.
11. **On-field dataset.** It is phrase-based and a floor; goals ruled out with other wording are missed.
12. **Pipeline defects.** A post-draft audit (Section 4.5, Appendix D) found five small defects (one non-VAR event counted as VAR, two weakly evidenced events, a foul-zone mis-merge, missing Understat aliases for two clubs and an abandoned match counted as 0–0). Their combined effect on any estimate is at most 0.05 on a rate ratio of about 2.0, but they illustrate that agreement between unofficial feeds does not rule out residual errors.

8.3 What would settle the question

- Semi-automated or tracking-based offside data (the position of the offside line at the moment of the pass) to test the runners-in-behind hypothesis directly.
- An official, event-level audit of every VAR intervention with a correct/incorrect label, which would give the error rate by beneficiary — the only direct test of favouritism.
- All VAR *checks* (not only overturns), to estimate overturn rates per review by team.
- The VAR-room official per match, to separate on-field and VAR-room effects.

9. Conclusions

1. **The post's direction is right, its numbers are not reproducible.** Real Madrid leads La Liga in VAR-nullified goals (50 vs 30), confirmed in two independent providers, but the post's counts (42 for Real Madrid) are lower than any full reconstruction and no single definition reproduces all seven teams.
2. **The outlier is real** (2.8× the rate of other teams; far outside the 99.8% funnel limit) and not an artefact of selection, clustering or data source.
3. **About half is explained** by how much Real Madrid attacks and by the offsides that come with it; a residual of roughly 2.0× remains after controlling for shots, xG, offsides, pressing, fouls and cards.
4. **The residual does not look like targeting or favouritism.** It is symmetric within matches, not concentrated in any referee, not specific to VAR (the same direction appears for on-field decisions), opponents are not spared, and VAR accuracy is too high for errors to produce it.
5. **The impact is modest.** 44 matches contain a nullified goal; in 13 the result would have changed (21 points in 8.2 seasons). Counting every nullified goal as wrong would change one title (2020-21); under realistic error rates the probability is at most about 5%.
6. **Open question.** The unexplained residual is most consistent with Real Madrid's attacking profile, a hypothesis that needs tracking data to confirm. We cannot exclude a modest VAR-specific effect (extra effect 1.36×, 95% CI 0.82–2.23), but we find no evidence of one.

> **In one sentence:** Real Madrid is a statistical outlier whose excess is roughly half explained by how it attacks and whose remainder shows none of the fingerprints of unequal treatment; the data neither prove nor require a bias explanation.

10. Reproducibility and data availability

All code, processed data, figures, the EDA notebook and the Excel report are public at `github.com/RoloMagnus/Real_Madrid_VAR_null_goals_outlier_study`. The pipeline scripts run in this order: `01_crawl_espn`, `02_crawl_flashscore`, `03_build_dataset`, `04_fetch_understat`, `06_referees_tm`, `05_enrich`, `09_fouls_espn`, `10_onfield_disallowed`, `11_onfield_events`, `07_analysis`, `08_build_excel`. The notebook `Real Madrid VAR - EDA.ipynb` reads the processed outputs and regenerates every figure and scenario table (it also runs without the raw caches). The raw caches are excluded from the repository because of size and provider terms; they are regenerated by the crawlers. Random seed: 2026.

References

1. Spitz J, Wagemans J, Memmert D, Williams AM, Helsen WF. Video assistant referees (VAR): The impact of technology on decision making in association football referees. *Journal of Sports Sciences* 2021; 39(2): 147–153. doi:10.1080/02640414.2020.1809163 (abstract verified via Europe PMC, PMID 32794432).
2. FIFA 2018 World Cup VAR statistics, as summarised in “Video assistant referee”, Wikipedia (accessed 12 Oct 2026): success rate 99.3% versus 95% without VAR; 335 incidents checked in the group stage, 14 overturned.
3. Sky Sports. “VAR: Premier League claims 96 per cent of referee decisions are correct”, 8 February 2024 (headline level; full text not retrieved).
4. BBC Sport. “VAR: Report shows technology has been 98.9% accurate in decision-making”, 22 January 2018 (headline level).
5. El País. “El arbitraje español sostiene que con el VAR acierta en el 98% de los penaltis”, 10 December 2020 (headline level).
6. MisterChip (@2010MisterChip). Post of 10 October 2026 on VAR-nullified goals in La Liga, 2018–2026 (image supplied by the study sponsor).
7. Understat, ESPN, Flashscore and Transfermarkt public pages and endpoints, queried 11–12 October 2026.

Appendix A. Event classification rules

- **Incident:** Flashscore *Goal Disallowed* (`IKX`) with reason code (`ILX`: offside, Foul, handball, video review), scorer, minute and side.
- **VAR-confirmed:** commentary icon `var` or text matching `VAR|video assistant|video review|disallow` within −1/+3 minutes of the incident, or an ESPN event of type *Deleted After Review*, a *VAR* decision text containing "No Goal", or a "Video Review" delay within three minutes.
- **Scorer vs build-up offside:** ESPN *Offside* event for the same team within ±1 minute; if the named player's token set equals or contains the scorer's, *goalscorer*; otherwise *teammate*; no match: *not identified*.
- **Points lost:** points(goals+1 for the team in that match) − actual points; multiple nullified goals in one match are added back together in the match-level analysis.
- **Half:** base minute ≤ 45 → first half, otherwise second (stoppage time inherits the half).

Appendix B. Supplementary statistics

Table 20. Full Poisson regression results (both scopes).

Scope	Model	RM IRR	CI low	CI high	p	Dispersion
All 20 teams per season	raw: RM indicator only	2.74	1.98	3.79	<0.001	1.07
All 20 teams per season	+ offsides	2.52	1.83	3.48	<0.001	1.03
All 20 teams per season	+ shots	2.20	1.57	3.07	<0.001	1.06
All 20 teams per season	+ shots on target	2.28	1.64	3.16	<0.001	1.06
All 20 teams per season	+ xG	2.24	1.61	3.11	<0.001	1.06
All 20 teams per season	+ offsides + shots	2.03	1.46	2.82	<0.001	1.03
All 20 teams per season	+ offsides + shots + xG + home + season FE	2.01	1.45	2.79	<0.001	1.03
Top-10 teams per season	raw: RM indicator only	2.37	1.68	3.32	<0.001	1.11
Top-10 teams per season	+ offsides	2.24	1.60	3.13	<0.001	1.06
Top-10 teams per season	+ shots	1.91	1.34	2.71	<0.001	1.10
Top-10 teams per season	+ shots on target	1.96	1.40	2.76	<0.001	1.10
Top-10 teams per season	+ xG	2.04	1.45	2.88	<0.001	1.09
Top-10 teams per season	+ offsides + shots	1.80	1.27	2.54	<0.001	1.07
Top-10 teams per season	+ offsides + shots + xG + home + season FE	1.84	1.30	2.60	<0.001	1.06

Table 21. Correlations between nullified goals and attack-volume variables across top-10 team-seasons (8 full seasons; bootstrap 95% CI).

Scope	Variable	n	Pearson r	CI low	CI high	p	Spearman rho
Top-10 team-seasons	Shots (chances created)	80	0.48	0.30	0.63	<0.001	0.45
Top-10 team-seasons excl. RM	Shots (chances created)	72	0.30	0.11	0.47	0.011	0.32
Top-10 team-seasons	Shots on target	80	0.44	0.25	0.60	<0.001	0.46
Top-10 team-seasons excl. RM	Shots on target	72	0.29	0.09	0.49	0.014	0.35
Top-10 team-seasons	xG	80	0.49	0.33	0.63	<0.001	0.51
Top-10 team-seasons excl. RM	xG	72	0.38	0.19	0.55	<0.001	0.43
Top-10 team-seasons	Offsides (attacking positions)	80	0.21	0.03	0.40	0.060	0.30
Top-10 team-seasons excl. RM	Offsides (attacking positions)	72	0.20	0.00	0.45	0.087	0.28
Top-10 team-seasons	Corners	80	0.37	0.19	0.54	<0.001	0.38
Top-10 team-seasons excl. RM	Corners	72	0.25	0.07	0.46	0.033	0.28
Top-10 team-seasons	Goals scored	80	0.44	0.27	0.59	<0.001	0.50
Top-10 team-seasons excl. RM	Goals scored	72	0.36	0.16	0.54	0.002	0.42

Appendix C. The 50 Real Madrid VAR-nullified goals

Table 22. Breakdown of Real Madrid VAR-nullified goals (round, venue H/A, score and result from Real Madrid's perspective, league position at kick-off). The full dataset with 28 fields, the manual validation flag and the video links is in the Excel report (sheet 2_RM_Nullified_Goals).

#	Date	Rd	Rival	Scorer	Min	Reason	Context	V	Score	Result	Pos.	Referee
1	2018-09-26	6	Sevilla	Luka Modric	54	Offside	Offside - goalscorer	A	0-3	LOSS	2	Alejandro Hernández Hernández
2	2018-10-20	9	Levante	Marco Asensio	18	Offside	Offside - player not identified	H	1-2	LOSS	5	Guillermo Cuadra Fernández
3	2019-03-16	28	Celta de Vigo	Luka Modric	56	Offside	Offside - goalscorer	H	2-0	WIN	3	Juan Martínez Munuera
4	2019-09-25	6	Osasuna	Luka Jovic	58	Offside	Offside - goalscorer	H	2-0	WIN	4	Mateu Lahoz
5	2019-11-02	12	Real Betis	Eden Hazard	8	Offside	Offside - goalscorer	H	0-0	DRAW	2	José María Sánchez Martínez
6	2020-01-26	21	Real Valladolid	Casemiro	12	Offside	Offside - player not identified	A	1-0	WIN	2	De Burgos Bengoetxea
7	2020-02-16	24	Celta de Vigo	Sergio Ramos	49	Offside	Offside - goalscorer	H	2-2	DRAW	1	Javier Alberola Rojas
8	2020-07-16	37	Villarreal	Marco Asensio	96	Handball	Handball (attacking team)	H	2-1	WIN	1	Alejandro Hernández Hernández
9	2020-10-04	5	Levante	Sergio Ramos	69	Video review (unspecified)	Offside - player not identified	A	2-0	WIN	4	José Luis Munuera Montero
10	2021-02-14	23	Valencia	Ferland Mendy	62	Offside	Offside - goalscorer	H	2-0	WIN	3	José María Sánchez Martínez
11	2021-04-03	29	Eibar	Marco Asensio	38	Offside	Offside - player not identified	H	2-0	WIN	3	Isidro Díaz de Mera Escuderos
12	2021-05-09	35	Sevilla	Karim Benzema	12	Offside	Offside - teammate in build-up	H	2-2	DRAW	3	Juan Martínez Munuera
13	2021-05-22	38	Villarreal	Karim Benzema	58	Offside	Offside - player not identified	H	2-1	WIN	2	José Luis Munuera Montero
14	2021-09-22	6	Mallorca	Karim Benzema	49	Foul	Attacking foul (before goal)	H	6-1	WIN	2	Javier Alberola Rojas
15	2022-02-26	26	Rayo Vallecano	Casemiro	39	Offside	Offside - goalscorer	A	1-0	WIN	1	Isidro Díaz de Mera Escuderos
16	2022-03-05	27	Real Sociedad	Karim Benzema	69	Offside	Offside - goalscorer	H	4-1	WIN	1	Jesús Gil Manzano
17	2022-04-17	32	Sevilla	Vinícius Júnior	76	Handball	Handball (attacking team)	A	3-2	WIN	1	Guillermo Cuadra Fernández
18	2022-04-30	34	Espanyol	Isco	72	Offside	Offside - goalscorer	H	4-0	WIN	1	José Luis Munuera Montero
19	2022-10-02	7	Osasuna	Karim Benzema	81	Offside	Offside - goalscorer	H	1-1	DRAW	2	Guillermo Cuadra Fernández
20	2022-10-08	8	Getafe	Rodrygo	57	Offside	Offside - goalscorer	A	1-0	WIN	2	Mateu Lahoz
21	2022-10-19	10	Elche	Karim Benzema	61	Offside	Offside - teammate in build-up	A	3-0	WIN	1	Jesús Gil Manzano
22	2022-10-19	10	Elche	David Alaba	26	Offside	Offside - goalscorer	A	3-0	WIN	1	Jesús Gil Manzano
23	2022-10-19	10	Elche	Karim Benzema	6	Offside	Offside - teammate in build-up	A	3-0	WIN	1	Jesús Gil Manzano
24	2022-10-30	12	Girona	Rodrygo	90	Video review (unspecified)	Other / unspecified VAR review	H	1-1	DRAW	2	Mario Melero López

#	Date	Rd	Rival	Scorer	Min	Reason	Context	V	Score	Result	Pos.	Referee
25	2023-02-02	17	Valencia	Antonio Rüdiger	48	Foul	Attacking foul (before goal)	H	2-0	WIN	2	Javier Alberola Rojas
26	2023-02-18	22	Osasuna	Vinícius Júnior	89	Offside	Offside - teammate in build-up	A	2-0	WIN	2	José Luis Munuera Montero
27	2023-03-05	24	Real Betis	Karim Benzema	12	Handball	Handball (attacking team)	A	0-0	DRAW	2	César Soto Grado
28	2023-03-19	26	FC Barcelona	Marco Asensio	81	Offside	Offside - goalscorer	A	1-2	LOSS	2	De Burgos Bengoetxea
29	2023-04-02	27	Real Valladolid	Rodrygo	66	Handball	Handball (attacking team)	H	6-0	WIN	2	Juan Luis Pulido Santana
30	2023-08-19	2	Almería	Toni Kroos	50	Foul	Attacking foul (before goal)	A	3-1	WIN	5	José María Sánchez Martínez
31	2023-10-21	10	Sevilla	Federico Valverde	4	Offside	Offside - goalscorer	A	1-1	DRAW	1	De Burgos Bengoetxea
32	2024-02-25	26	Sevilla	Lucas	10	Foul	Attacking foul (before goal)	H	1-0	WIN	1	Isidro Díaz de Mera Escuderos
33	2024-10-26	11	FC Barcelona	Kylian Mbappé	30	Offside	Offside - goalscorer	H	0-4	LOSS	2	José María Sánchez Martínez
34	2024-12-04	19	Athletic Club	Kylian Mbappé	14	Offside	Offside - goalscorer	A	1-2	LOSS	2	José María Sánchez Martínez
35	2025-01-03	12	Valencia	Kylian Mbappé	60	Offside	Offside - goalscorer	A	2-1	WIN	2	César Soto Grado
36	2025-01-19	20	Las Palmas	Kylian Mbappé	42	Offside	Offside - goalscorer	H	4-1	WIN	2	Alejandro Quintero González
37	2025-01-19	20	Las Palmas	Jude Bellingham	73	Offside	Offside - teammate in build-up	H	4-1	WIN	2	Alejandro Quintero González
38	2025-01-19	20	Las Palmas	Federico Valverde	86	Offside	Offside - teammate in build-up	H	4-1	WIN	2	Alejandro Quintero González
39	2025-04-05	30	Valencia	Mouctar Diakhaby	21	Offside	Offside - teammate in build-up	H	1-2	LOSS	2	Guillermo Cuadra Fernández
40	2025-04-13	31	Deportivo Alavés	Raúl Asencio	19	Foul	Attacking foul (before goal)	A	1-0	WIN	2	César Soto Grado
41	2025-04-20	32	Athletic Club	Vinícius Júnior	79	Offside	Offside - teammate in build-up	H	1-0	WIN	2	Juan Martínez Munuera
42	2025-08-30	3	Mallorca	Arda Güler	55	Handball	Handball (attacking team)	H	2-1	WIN	3	José María Sánchez Martínez
43	2025-08-30	3	Mallorca	Kylian Mbappé	46	Offside	Offside - goalscorer	H	2-1	WIN	3	José María Sánchez Martínez
44	2025-08-30	3	Mallorca	Kylian Mbappé	7	Offside	Offside - goalscorer	H	2-1	WIN	3	José María Sánchez Martínez
45	2025-09-13	4	Real Sociedad	Arda Güler	2	Offside	Offside - teammate in build-up	A	2-1	WIN	1	Jesús Gil Manzano
46	2025-10-26	10	FC Barcelona	Kylian Mbappé	12	Offside	Offside - goalscorer	H	2-1	WIN	1	César Soto Grado
47	2025-11-30	14	Girona	Kylian Mbappé	40	Handball	Handball (attacking team)	A	1-1	DRAW	3	De Burgos Bengoetxea
48	2026-08-26	1	Real Sociedad	Brahim Díaz	90	Offside	Offside - teammate in build-up	H	4-1	WIN	7	Mateo Busquets Ferrer
49	2026-09-04	4	Real Betis	Carlos Espí	87	Offside	Offside - goalscorer	A	0-1	LOSS	2	Alejandro Hernández Hernández
50	2026-10-10	8	Villarreal	Vinícius Júnior	71	Foul	Attacking foul (before goal)	H	1-0	WIN	4	Isidro Díaz de Mera Escuderos

Appendix D. Pipeline audit (15 October 2026)

Summary of the audit described in Section 4.5; scripts in `scripts/audit/`, details in `AUDIT.md`. Defects were tested in an isolated copy of the pipeline; the published scripts, data and estimates are unchanged.

Table 23. Audit checks and results.

Check	What was tested	Result
Minute	Flashscore incident minute vs ESPN explicit-VAR minute (286 events)	Identical 47%; within 1 min 79%; within 2 min 95.5%; all inside a +/-3 min window; Flashscore earlier in 136 (ESPN logs the decision)
VAR evidence	Independent re-derivation of evidence for 397 candidate incidents	286 ESPN explicit + Flashscore wording; 105 Flashscore wording only; 2 ESPN video-review delay only; 4 no VAR reference
Text relevance	VAR commentary in the window concerns the goal (not a penalty/card)	389 of 389 (also under a stricter pattern)
Double counting	Two incidents, same team and match, within 2 minutes	0
Recall	ESPN "overturned by VAR" events 2022-23 onwards present in our list	122 of 128; none of the 6 missing involves Real Madrid
ESPN merge	Team names agree for single-candidate matches (kick-off hour + score key)	3,034 of 3,034; 78 ambiguous left blank (18 events); 6 without candidate
Understat merge	Flashscore goals vs Understat goals scored/conceded	6,120 of 6,120 agree; 116 not joined (Huesca 76, Oviedo 38, abandoned match 2)
Referee merge	Transfermarkt full name vs Flashscore short name, 617 matches	616 of 617 surnames agree
Foul-zone merge	Season + home + away key, old vs corrected name mapping	1 mis-merged match; 7 matches without foul-zone data
Hand-built sample	Pipeline output for the four sample rows	Correct for all four
Manual validation file	Rows vs matches vs video links	50 rows, 44 matches, 44 distinct links; one link per match, none shared

Table 24. Defects found, effect and status.

#	Defect	Effect	Status
1	Bare word "disallow" accepted as VAR evidence	One non-VAR event counted (Getafe 2020): 394 instead of 393; Real Madrid unchanged	Documented, not applied
2	Two events counted on an ESPN video-review delay only	Includes Real Madrid-Levante 2018 (video-validated); strict rule: 391 events, Real Madrid 49, rate ratio 2.72	Kept; sensitivity reported
3	Substring name fallback maps ESPN "Deportivo" to Deportivo Alaves (foul-zone script, ESPN check)	1 match received another match's foul data; 7 La Coruna matches without foul zones	Documented, not applied
4	Understat aliases missing for Huesca and Oviedo	xG missing for 76 team-matches, PPDA for 114 (+ opponents' terms): 230 team-matches leave the PPDA models	Documented, not applied
5	Abandoned match (Rayo Vallecano-Athletic, 10 Oct 2026) counted as 0-0	2 team-matches; no nullified goal; marginal effect on the partial 2026-27 table	Assumption, documented

Table 25. Impact of fixing defects 1, 3 and 4 on the headline estimates (isolated copy; conclusions unchanged).

Estimate	Published	With defects 1, 3 and 4 fixed (isolated copy)
VAR-confirmed events / Real Madrid	394 / 50	393 / 50
Rate ratio per match, all teams (top-10)	2.76 (2.37)	2.77 (2.38)
Volume-model IRR, all teams (n)	2.01 (6,160)	2.04 (6,236)
Volume-model IRR, top-10	1.84	1.85
+ PPDA, all teams (n)	1.99 (6,006)	1.94 (6,234)
+ fouls, cards and PPDA, all teams (n)	2.02 (5,980)	1.97 (6,224)
Leave-Real-Madrid-out forecast (95% interval)	25.0 (16-35)	24.6 (15-35)
Within-match symmetry p (offsides / goal opportunities)	0.110 / 0.135	0.110 / 0.130
Opponents spared: obs/exp (p)	0.70 (0.112)	0.70 (0.114)
Referee heterogeneity p	0.617	0.617
Extra Real Madrid effect at the VAR stage (p)	1.36 (0.23)	1.35 (0.24)
Net points ledger, Real Madrid	-9	-9
xG / PPDA missing	1.22% / 1.86%	0.00% / 0.03%